

You work for a data analytics company that uses the Google Cloud infrastructure for performing all its data warehousing and business intelligence workloads. You need to perform an evaluation of the cloud infrastructure and investigate uptimes and response rates. Which Google Cloud tool should you use for this?

- A) Cloud Vision

✖
- B) Cloud Monitoring

✔
- C) Cloud Debugger

✖
- D) Cloud Trace

✖

Explanation

You should use Cloud Monitoring for this scenario. Cloud Monitoring can be used to monitor the complete infrastructure of a cloud deployment. It provides a view into the uptime, health, and performance of cloud systems. For this, Cloud Monitoring gathers relevant metadata, events, and metrics from sources that include uptime probes, services, app instrumentation, alerts, notifications, and other components of applications. It allows you to stop issues from happening, and it allows you to identify trends. It also reduces overhead related to monitoring through a service that is integrated and provides metrics, dashboards, and alerts all in one place. Cloud Monitoring provides service level objective (SLO) monitoring and metrics collection for virtual machines (VMs) (using Ops Agent) and Kubernetes (using Managed Service for Prometheus).

You would not use Cloud Trace for this scenario. Cloud Trace is a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights into its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

You would not use Cloud Vision for this scenario. Cloud Vision is used by developers to understand image content through machine learning (ML) models accessed through an application programming interface (API). It provides image classification into various categories, object and face detection inside images, and identification of printed words inside of images.

You would not use Cloud Debugger for this scenario. Cloud Debugger is better suited for performing an analysis of an application's state at a specific location of code without the need to halt or slow down application requests. Cloud Debugger does this by creating a connection between the production data of your application to the source code.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Identify Google Cloud solutions for cloud resource monitoring and application performance management

References:
[Cloud.google.com > Cloud Monitoring](#)

Related Material

A company that develops database applications is looking for a Google Cloud service that will enable its applications to get better performance by incorporating the availability and scalability of Redis services without the associated complexities of managing the Redis deployments. Which service should you use in this scenario?

- A) Memorystore

✔
- B) Datastore

✖
- C) Firestore

✖
- D) Cloud Bigtable

✖

Explanation

You should use Memorystore in this scenario. Memorystore for Redis is a scalable, fully managed service for Redis on Google Cloud that provides 99.9% availability using up to five read replicas that are distributed across multiple availability zones (AZs). Memorystore provides automation for complex Redis and Memcached tasks, which include ensuring high availability, creating failovers, and performing patching. It is compatible with the protocols of both Memcached and Redis. Memorystore also has provision for private IP addresses and virtual private cloud networks (VPCs) for data security and has integration with Identity and Access Management (IAM) for access control. Memcached provides in-memory caching for temporary data storage. Redis is a key-value NoSQL database using an in-memory data system for high performance. Redis supports sharding and uses persistent data, while Memcached does not.

Firestore is not the appropriate choice for this scenario. Firestore is a serverless document database that is scalable and fully managed. It enables fast development for applications for the Internet, mobile, and the Internet of Things (IoT) that can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features. It also allows you to customize various rules for security and data validation to protect your data. Firestore can be integrated with BigQuery and Firebase. Firestore is an upgrade to the older service called Datastore, which provides a NoSQL database that is scalable.

Dataprep is not the appropriate choice for this scenario. Dataprep is a data service for exploring, cleaning, and preparing data visually for the purpose of machine learning (ML) and analysis. It is an intelligent data service that is serverless, so the user does not need to deploy or manage the infrastructure. Dataprep suggests the data transformations that are ideal each time a User Interface (UI) input is detected, so the user does not need to write code for these applications. Dataprep also detects any anomalies that may exist in the data, including duplicates and missing value data. It helps the user avoid having to assess the quality of the data and proceed directly to exploratory data analysis.

Cloud Bigtable is not the appropriate choice for this scenario. Cloud Bigtable is a NoSQL database service that is fully managed and enables you to work with massive workloads for analytics and operations with availability at 99.999%. It is ideal for uses that include applications for IoT, finance, banking, personalized services, and advertising technology. Cloud Bigtable has a storage engine that allows for improved predictions in applications utilizing ML. It also has integration with tools for big data including Hadoop, Dataproc, and Dataflow.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > Memorystore](#)

Related Material

You are creating a plan for digitally transforming workloads for your company. What would be the purpose of using cloud-native applications during application modernization?

- A) Consolidating a single unified backend for redundancy

✖
- B) Using APIs for ensuring all business teams work at the same pace

✖
- C) Partitioning workloads for failure protection

✔
- D) Avoiding cost overruns caused by application decoupling

✖

Explanation

Using cloud-native applications allows for partitioning workloads for failure protection. The process of functional decomposition involves identifying individual components and their related systems inside of existing workloads. Then, you can create a decoupled architecture where workloads are broken into smaller parts and scaled independently. This can help you provide higher availability of services as well as protection from failures by maintaining several read replicas of a service within a system that is distributed.

Consolidating a single unified backend for redundancy is not a purpose of application modernization. You would instead spread backends and other services over several availability zones (AZs) and use read replicas and backups to ensure that the system is not affected by failures. You may need to maintain certain services in a specific geographical location for compliance, and you may need to distribute other services globally for low latency content distribution to users.

Avoiding cost overruns caused by application decoupling is not a purpose of application modernization. Performing decoupling of applications as part of application modernization ensures that services that scale up or down more frequently are separated from other services that are more stable. This way, you cut back on costs associated with scaling up of services that are not required or are unrelated.

Using application programming interfaces (APIs) to ensure all business teams work at the same pace is not a purpose of application modernization. You would use APIs to perform application decoupling for existing business services. This is a form of functional decomposition that breaks down an existing legacy or “monolith” application into independent applications. With APIs for various business services, multiple service users and teams can use these services at their own pace. This system also provides greater visibility into the usage of these services in the company.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand modernizing applications with Google Cloud

References:
[Inthecloud.withgoogle.com > CIO Guide to Application Modernization](#)

Related Material

Question #4 of 60

Question ID: 1497151

A company's cloud deployment has already had a lot of budget overruns, and the company needs to cut back on cloud expenditure immediately. You need to implement financial governance strategies to gain more control over cloud costs. How would you achieve this?

- A) Providing cost viewing to all cloud users

✖
- B) Ensuring each user can provision cloud resources

✖
- C) Using organizational policies for resources

✔
- D) Decommissioning unused on-premises servers

✖

Explanation

You would use organizational policies for resources to gain more control over your cloud costs. The purpose of using organizational policies is to specify permissions for users to perform spending on Google Cloud based on their role in the organizational hierarchy. This way, you control the ability of specific users to view costs and perform administrative functions using Cloud Identity and Access Management (Cloud IAM) for cloud resources. You will also use cost management controls to ensure you don't spend beyond your stipulated budget and plan costs properly. You can set limitations on cost accumulations using quotas and perform monitoring for usage and costs using budgets. You can perform automated throttling and capping of resource consumption as well as broadcast budget notification alerts on company communication channels, such as Slack.

Providing cost viewing to all cloud users is not the correct way to gain more control over your cloud costs because each user on a cloud deployment cannot be allowed more privileges than their job role requires, as this can cause security as well as cost management issues. Cloud resource usage and cost viewing permissions need to be made based on the specific role a user performs in the organization.

Ensuring each user can provision cloud resources is not the correct way to gain more control over your cloud costs because allowing all users to provision cloud resources can cause major cost overruns and cost control issues. Correct financial governance on the cloud is to allow resource provisioning based on the user's role and position in the organizational hierarchy.

Decommissioning unused on-premises servers is not the correct way to gain more control over your cloud costs because the state of on-premises infrastructure does not affect cloud operating costs as the two systems are disparate. Decommissioning unused servers from an on-premises data center can help improve the operational effectiveness of an on-premises data center; However, this will not have an effect on how resources are provisioned on the cloud deployment, which requires a different system be in place to ensure cost control, such as using IAM policies.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > A guide to financial governance in the cloud](#)

Related Material

Question #5 of 60

Question ID: 1497145

You are in the process of creating a working environment for developers in your various machine learning teams. Which of these options would you use for a development environment that is based on a Jupyter notebook and allows for data processing using a Dataproc cluster?

- A) CCAI

✖
- B) Firestore

✖
- C) Vertex AI Workbench

✔
- D) Dialogflow

✖

Explanation

You would use Vertex Artificial Intelligence (AI) Workbench for this scenario. This provides a development environment that is based on the Jupyter notebook system and can be utilized for the entire workflow you use for your data science tasks. With Vertex AI Workbench, you can work with data in Dataproc clusters, perform model training, and propagate the results from within the interface of JupyterLab. Vertex AI also contains data labeling jobs that you can use to perform data labeling for datasets that will be used for model training. The Vertex AI Feature Store provides a repository that is fully managed for ingestion, serving, and sharing feature values of ML inside of a company.

Dialogflow is not the appropriate choice for this scenario. Dialogflow provides a suite for the development of conversational apps with voice and text capabilities, which include voicebots as well as chatbots. Dialogflow provides cross-platform connectivity to existing apps for the Internet of Things (IoT), iOS, or Android, and for telephony platforms, including Cisco, Genesys, and Avaya. It also can be used for digital platforms, including Slack, Facebook Messenger, and Google. Dialogflow Essentials (ES) is the standard edition, and Dialogflow Customer Experience Edition (CX) is the advanced edition with features like flow visualizations that are interactive and advanced AI features.

Firestore is not the appropriate choice for this scenario. Firestore is a serverless document database that is scalable and fully managed. It enables fast development for applications for the internet, mobile, and the Internet of Things (IoT) that can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features.

Contact Center Artificial Intelligence (CCAI) is not the appropriate choice for this scenario. CCAI is used for enhancing the customer service experience for your customers in existing contact centers by leveraging AI technology. CCAI includes Dialogflow CX, Dialogflow Essentials, Speaker ID, Text-to-Speech, and Speech-to-Text. The CCAI Platform is a Contact Center as a Service (CCaaS) platform that is AI enabled and created on Google Cloud natively. It is built to be used with Customer Relationship Management (CRM) systems and provides a unified source of data for customer experiences. Contact Center AI Insights enables customers to perform value extraction from their data in contact centers. They can use Contact Center AI Insights to search for key information and perform necessary actions based on that data using a console. Customers can also use Contact Center AI Insights for extraction of topics, key areas, and sentiments from data using advanced analytics.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > Introduction to Vertex AI](#)

Related Material

Question #6 of 60

Question ID: 1497406

Your operations and development teams need to create a plan for ensuring service availability while adding some key enhancements. For this purpose, which of these can you consider valid service level indicators (SLIs) to meet your service level objectives (SLOs)? (Choose two.)

- A) Number of successful login attempts**

✔
- B) Number of API calls generated for the service**

✘
- C) Amount of memory consumed by the service**

✘
- D) Percentage of correct user responses**

✔
- E) CPU utilization for the service**

✘

Explanation

The percentage of correct user responses and the number of successful login attempts are both valid service level indicators (SLIs) that you can use to meet your service level objectives (SLOs). SLIs measure the quantity of a specific aspect of the service that is relevant to the user. SLOs are defined based on user experience and can address user-centric needs, such as latency. SLIs help gauge if SLOs are met. A combination of SLOs and SLIs work towards meeting service level agreements (SLAs), which specify the exact agreement made with the users of the service.

CPU utilization for the service, the amount of memory consumed by the service, and the number of application programming interface (API) calls generated for the service are all not the best options for SLIs as these are internal to the system and not user-centric. By selecting SLIs that capture performance aspects that are relevant to the user, you can better meet SLA needs.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Explain how IT operations need to adapt to thrive in the cloud

References:
[Services.google.com > Increasing Business Value with Better IT Operations: A guide to Site Reliability Engineering \(SRE\)](#)

Related Material

Question #7 of 60

Question ID: 1495162

You work for a company that is considering revising its plan of procuring physical servers and infrastructure to using the Google Cloud platform instead.

Which of these should you suggest as a key benefit of using Google's Cloud computing platform?

- A) Quick deployment of new applications without provisioning underlying infrastructure**

✔
- B) Investment by the company in the latest computing technologies**

✘
- C) A pricing system that ensures both used and unused resources are paid for**

✘
- D) Access of compute services exclusively through an on-premises networking system**

✘

Explanation

You should suggest quick deployment of new applications without provisioning underlying infrastructure as a key benefit of using Google's Cloud computing platform. This is because the infrastructure required for developing and deploying applications is managed by Google Cloud, which frees up a company to use its staff for creating applications needed to drive business growth.

Access of compute services exclusively through an on-premises networking system is not a key benefit of using Google's Cloud computing platform. This is because when using a cloud computing system, users can access computing resources using an Internet connection from anywhere.

Investment by the company in the latest computing technologies is not a key benefit of using Google's Cloud computing platform. The company does not need to make investments in new and emerging technologies because it is the cloud providers' expertise to provide the necessary services to the company using the latest technologies. This saves the company a lot of capital by not having to invest in trending technologies that may be out of use in a few years.

A pricing system that ensures both used and unused resources are paid for is not a key benefit of using Google's Cloud computing platform. This is because in cloud computing, a company only needs to pay for the resources it actually consumes and not for any unused resources. This is also referred to as a Pay-As-You-Go (PAYG) system.

The key benefits of cloud computing you need to know about include:

- Flexibility – Cloud computing is architected in a way that allows users to access cloud systems from any location using an Internet connection and have scalability for services as per their requirements.
- Efficiency – Cloud computing allows companies to deploy new applications quickly without needing to take on the overhead of the necessary infrastructure.
- Strategic value – Cloud service providers have access to the latest technologies, which customers can avail of as services. This removes the risk of clients having to invest in new technologies which may become obsolete in time.
- Security – Security risks on a cloud platform are relatively low compared to an equivalent data center for an enterprise. This is due to the strength of cloud security systems and teams.
- Cost effectiveness – Cloud computing ensures that companies only pay for resources consumed. This ensures that data centers do not need to be provisioned with more capacity to handle spikes in usage or organizational growth. This allows a company to have its staff work on strategic tasks which help grow the business.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why cloud technology is revolutionizing business

References:
[Cloud.google.com > What is Cloud Computing? > What are the benefits of cloud computing?](#)

Related Material

Question #8 of 60

Question ID: 1495901

You need to suggest a scaling strategy for a company that wants to move from its maturity level of strategic cloud adoption to one that is transformational. Which of these actions will you suggest for this requirement?

- A) Perform each modification when issues are found
- B) **Perform each modification using a programmatic approach**
- C) Perform each modification using a manual approach
- D) Keep the system immutable with no modifications
- ✖

✔

✖

✖

Explanation

You will suggest that the company perform each modification using a programmatic approach. This is because taking a transformational approach involves the use of automation. Automation ensures that changes are made in a constant and consistent manner with low risks to the system. This also ensures that changes can be rolled back, if necessary, to maintain a safe system configuration.

You would not perform each modification using a manual approach. This is not a transformational approach, but a tactical approach that involves changes being made infrequently and can pose high risks for the company. This approach is not recommended.

You would not perform each modification when issues are found. This is not a transformational approach that ensures that risks and problems are mitigated before they actually occur.

You would not keep the system immutable with no modifications. This is because a transformational approach requires rethinking and redefining existing systems and relationships for leveraging new technologies to facilitate operational excellence and business growth.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why it is critical for businesses to adopt new technology

References:
[Services.google.com > The Google Cloud Adoption Framework](#)

Related Material

Question #9 of 60

Question ID: 1495292

Your company has various systems and services running on Google Cloud. Each of these generates events. You need a way to create alerts whenever a specific error event on the system is generated. Which Google Cloud service should you use to accomplish this?

- A) Cloud Vision
- B) Cloud Spanner
- C) Cloud Armor
- D) **Cloud Logging**
- ✖

✖

✖

✔

Explanation

You should use Cloud Logging for this scenario. Cloud Logging is a log management service that is fully managed. It allows you to store, search, and analyze exabytes of log data, as well as set up alerts and notifications. Cloud Logging has Logs Explorer, which lets you use flexible queries for searching, sorting, and analyzing logs. You can also utilize log buckets to store and process log data based on regional compliance requirements. Cloud Logging also has an Error Reporting feature that performs a smart collection of errors in logs into error groups. Cloud Logging also allows your security teams to keep audit trails using its Cloud Audit Logs feature, which provides greater transparency in activities related to administration. Cloud Logging uses the Cloud Logging Application Programming Interface (API) to collect log entries passed through a Logs Router. The Logs Router uses existing filters to identify the log entries meant for exclusion, ingestion, or importing.

You would not use Cloud Spanner for this scenario. Cloud Spanner is a relational database service that is fully managed and ideal for critical business applications. You would use Cloud Spanner for the provision of a highly scalable database for online transaction processing (OLTP). This database will have high levels of availability and consistency worldwide.

You would not use Cloud Vision for this scenario. Cloud Vision is used by developers to understand image content through machine learning (ML) models accessed through an API. It provides image classification into various categories, object and face detection inside images, and identification of printed words inside of images.

You would not use Cloud Armor for this scenario. Cloud Armor is a service that protects websites and software applications from threats like web attacks and distributed denial of service (DDoS) attacks. It offers adaptive protection by detecting and mitigating large numbers of Layer 7 DDoS incidents using a locally trained machine learning (ML) system.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Identify Google Cloud solutions for cloud resource monitoring and application performance management

References:
[Cloud.google.com > Cloud Logging](#)

Related Material

Question #10 of 60

Question ID: 1495896

Which of these definitions best describes the meaning of the key term “Launch and iterate” in the context of revolutionizing business using Google’s cloud technology?

- A)** A focus on creating solutions that make improvements by order of 10 times
- B)** A focus on creating a solution that makes improvements by 10 percent
- C)** A focus on building a perfect solution right at the very beginning
- D) A focus on finding the right solution through continuous experimentation**

Explanation

The meaning of the key term “Launch and iterate” is best described as “A focus on finding the right solution through continuous experimentation”. This approach encourages organizations to constantly try out new systems and technologies to arrive at a solution that works rather than attempting to create a perfect solution from the start.

“A focus on building a perfect solution right at the very beginning” is not the meaning of launch and iterate. Finding the perfect solution at the start is not recommended because continuous experimentation with new technologies can help create the right solutions.

“A focus on creating solutions that make improvements by order of 10 times” and “A focus on creating a solution that makes improvements by 10 percent” are both incorrect. “Think 10x” is a key term that focuses on creating solutions that make improvements 10 times over.

For the Google Cloud Digital Leader exam, you need to pay special attention to definitions of certain key terms used across the exam. Your correct answer choice will depend greatly on your familiarity with these terms. Going through the associated reference material is also recommended.

- Cloud – This comprises a data center network that provides storage and computation for information using the Internet. The cloud includes all the systems that make this possible, including computers, software applications, networking systems, and security technology.
- Cloud technology or cloud computing – This refers to the systems and procedures required for storing, managing, and providing access to data sent across the cloud. This data is separate from the data that resides on a user’s personal computer’s hard drive.
- Data – In the context of Google Cloud, data refers to information that is of value to a company. Data can include audio and video content, textual information in an email, numbers in a spreadsheet file, imagery, and employee ideas. Data can include information that is internal or external to a company.
- Digital transformation – This occurs when a company utilizes new technology to change the design and definition of its relationships with its partners, customers, and employees. Digital transformation involves using new digital systems including private, public, and hybrid clouds to change existing business culture, work processes, and customer experiences with an aim to meet the changing dynamics of businesses and markets.
- Computing – This is the ability of a machine to perform storage, processing, retrieval, comparisons, and analysis of information. It also includes the machine's ability to automate tasks using computer applications.
- Compute power – This refers to the speed of a computer in processing data.
- Think 10x – This is also referred to as “generating big ideas” and involves looking for solutions to make improvements by an order of 10 times rather than by 10 percent.
- Launch and iterate – This is also referred to as “continuous learning” and involves a perspective of finding the right solution through continuous experimentation rather than attempting to craft a perfect system right at the beginning.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why cloud technology is revolutionizing business

References:
[Service.google.com > Cloud Digital Leader Study Guide](#)

Related Material

Question #11 of 60

Question ID: 1497143

You have an existing ML workflow, and you need to investigate how Vertex AI can be used in it. Which of these stages in your ML workflow can you manage using Vertex AI? (Choose two.)

- A)** Data lake modernization
- B)** Asynchronous messaging between applications
- C)** System bottleneck identification
- D) Dataset creation and data upload**
- E) ML model training and evaluation**

Explanation

You can manage Dataset creation, data upload, and ML model training and evaluation using Vertex Artificial Intelligence (AI). Vertex AI provides a singular application programming interface (API) platform to build and scale ML models swiftly with features for custom tooling and pre-training. It allows you to manage the AI and ML development lifecycle for a company. These stages of the ML workflow can be managed using Vertex AI:

- Dataset creation and data upload
- ML model training and evaluation
- Hyperparameter tuning
- Uploading and storing of ML models inside Vertex AI
- Deployment of the trained model onto an endpoint for prediction serving
- Sending of prediction requests to the endpoint and setting traffic splits
- Performing management of models and endpoints

You would not use Vertex AI for data lake modernization. You can perform data lake modernization, Extract, Transform, and Load (ETL) operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not use Vertex AI for system bottleneck identification. You can use Cloud Trace for such applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance.

You would not use Vertex AI for asynchronous messaging between applications. You would use Pub/Sub for such applications. Pub/Sub is a middleware for messaging that enables applications to send messages to each other asynchronously. An application designated as a “publisher” application sends messages to a topic, and all the applications that have subscribed to that topic will receive those messages.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > Introduction to Vertex AI](#)

Related Material

Question #12 of 60

Question ID: 1495225

Your company has various data analytics workloads running on-premises. You are in the process of using a distributed deployment pattern to create a hybrid cloud environment for your company. Which of these actions should you take to create a hybrid cloud environment for analytical workloads?

- A) Use a public cloud for creating a data lake**

✔
- B) Perform dataset processing on a private cloud**

✘
- C) Load raw data from a public cloud to a private cloud**

✘
- D) Deploy analytical workloads on a private cloud**

✘

Explanation

You should use a public cloud for creating a data lake. Using cloud storage for creating data lakes to be used for analytical processing provides the necessary flexibility for a company to gather petabytes of relevant data efficiently and economically. A data lake allows you to store both structured, semi-structured, and unstructured data from multiple sources in a single place. This data can come from a variety of sources, including multimedia, social media, hard disks, servers, Internet of Things (IoT) devices, clickstream, databases, and transactional information. A data lake provides security and scalability and enables companies to perform ingestion of data from various sources, including edge computing, cloud, or on-premises systems.

You would not deploy analytical workloads on a private cloud because analytical workloads require large levels of data processing with bursting characteristics, which are benefits of public cloud services. Dynamic resource provisioning and scaling offered in public cloud systems ensures efficient processing of massive datasets without the large investments required for implementing similar solutions using a private data center approach.

You would not load raw data from a public cloud to a private cloud. Raw data first needs to be extracted from the workloads that are operating in a private cloud or on-premises system and then loaded for analytics processing onto a public cloud. The results of this processing can then be sent back to existing transactional applications as feedback.

You would not perform dataset processing on a private cloud. This is because a private cloud or an on-premises system does not have the elasticity and scalability required to perform large scale computational and storage tasks that are required for analytics processing. You would instead use a public cloud for processing massive datasets without the need to arrange resources for these tasks.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Hybrid and multi-cloud architecture patterns](#)

Related Material

Question #13 of 60

Question ID: 1495228

You need to guide your company toward using more cloud-native services to create greater business value. Which of these are characteristics of serverless computing using the cloud? (Choose two.)

- A) Suitable only for predictable scaling needs**

✘
- B) Computing resources need to be managed by users**

✘
- C) Infrastructure is provisioned by the cloud service provider**

✔
- D) Code is run by actual servers**

✔
- E) Operational costs are higher**

✘

Explanation

Two characteristics of serverless computing include:

- Infrastructure is provisioned and scaled by the cloud service provider (CSP)
- Code is run by actual servers

Serverless computing is a cloud computing model that allows you to create and execute code without needing to manage the underlying servers or pay for any infrastructure that is unused.

Some serverless Google technologies include:

- Application development – You can use Cloud Functions and App Engine for serverless hosting of applications. You can use Cloud Datastore and Firebase for NoSQL serverless databases.
- Analytics – You can use Cloud Storage and Pub/Sub for serverless data analytics work and Dataflow for processing global scale datasets.
- Microservices – Google serverless systems help develop microservices based systems by bringing down the cost of infrastructure by dynamic scaling. Using fully managed serverless systems also reduces operational costs, which removes the need for companies to perform server management. Also, using serverless systems like Pub/Sub can help avoid synchronous call chains by creating precomputed results asynchronously for microservices.

Computing resources do not need to be managed by users. This is done by the CSP and allows developers to focus on creating business value instead of performing resource provisioning. Also, with the automatic and dynamic nature of resource provisioning using serverless systems, applications developed on serverless systems can perform massive levels of scaling.

Serverless systems do not create higher operational costs. This is because when provisioning servers for database and computing applications is done by the CSP, the operational costs to provision and manage massive server clusters reduce substantially.

Serverless systems are suitable not only for predictable scaling needs but excellent for unpredictable workloads. Serverless systems like Google's Cloud Functions and App Engine both offer dynamic scaling based on request volumes without the need for performing server management by the developer. You can use Cloud Functions for executing small chunks of code triggered by Hypertext Transfer Protocol (HTTP) requests or cloud events. Since the billing is only done per request, this can result in significant savings for traffic patterns that are uneven or unpredictable. Similarly, you can use App Engine for larger sections of code like for a Website.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Serverless Architecture](#)

Related Material

Question #14 of 60

Question ID: 1495927

Your company wants to digitally transform its existing systems. Which of these are key business drivers for modernizing applications for your company? (Choose two.)

- A) Creation of more agile systems at optimized costs**
- B) Increasing system visibility and control over spread-out operations**
- C) Remove the need for data gathering from changing markets
- D) Increase staff productivity through greater change management
- E) Increasing the TCO for facilitating necessary growth
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Explanation

Increasing system visibility and control over spread-out operations and the creation of more agile systems at optimized costs are both key business drivers for modernizing applications. By modernizing applications and using cloud-native services, you can have granular control over your development and operational systems, which can be automated to ensure a robust, change-friendly system. With fully managed services and serverless technology, you can have more agile systems in place with lower costs due to lessened overhead for the company.

Increasing the total cost of ownership (TCO) for facilitating necessary growth is not a key business driver. The TCO would be significantly reduced when performing application modernization. This is because application modernization will require less investment by the company for leveraging the latest trending technologies as well as the ability to use managed services for greater scalability and agility. Application modernization ensures that less time and resources are spent on tasks like resource provisioning, scaling, backups, and version updating, which cuts down on the capital that would be required to implement similar functionalities on a legacy system.

Removing the need for data gathering from changing markets is not a key business driver. When modernizing applications, you are able to gather more data from both historical sources as well as trending markets using modern data gathering systems and tools for business intelligence (BI). Application modernization will enable you to use fully managed services for creating data lakes and data warehouses for big data analysis and use query services to deliver useful insights on the same for powering business decisions.

Increasing staff productivity through greater change management is not a key business driver. By using managed services and automation, you can ensure that staff members of varying technical skill levels can achieve greater productivity without the need for intensive change management. Managed artificial intelligence (AI) systems can ensure that even people without the necessary coding skills for creating machine learning systems can use cloud services to build systems that deliver business value by leveraging AI technology to better understand markets and customers.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand modernizing applications with Google Cloud

References:
[Inthecloud.withgoogle.com > CIO Guide to Application Modernization](#)

Related Material

Question #15 of 60

Question ID: 1495943

A digital imaging company using Google Cloud has asked you to calculate downtime for its image processing service. The service has an availability SLO of 99.5% for a month. What is the error budget of the service?

- A) 99%
- B) 0%
- C) 100%
- D) 0.5%
-
-
-
-

Explanation

The error budget in this scenario is 0.5%. This is equivalent to about three and a half hours of downtime. The error budget is the measure of the system’s possible downtime, which is the difference between the service level objective (SLO) and the total availability figure of 100%:

Error budget = 100% – SLO = 100% – 99.5% = 0.5%

When an SLO is agreed upon, its corresponding service downtime or error budget is also agreed upon. An SLO specifies a definite target for a service, which can be 99.999% uptime. SLOs are defined based on user experience and can address user-centric needs, such as latency.

99% is not the error budget for this scenario. The value of 99% can be acceptable as the service level agreement (SLA) value. You need to keep an SLO value that is higher than the SLA, which helps you get early warnings when there is a danger of not meeting the SLA. SLAs are visible to customers that use a service while SLOs and SLIs are internal to the development and operations team. If you do not meet an SLO, then there is a possibility of not meeting the SLA which can cause issues for the company.

100% is not the error budget for this scenario. The value of 100% can be the desired value for reliability for a system, but it is not achievable in a real-world scenario. Costs and reliability need to be balanced in all real-world applications, and a system always depends on other systems, which reduces the level of reliability because the failure of one system can have an effect on others.

0% is not the error budget for this scenario. There is always a measurable value for an error budget because no system can have 100% uptime, and the error budget specifies the possible downtime that the system can face. When new features are added to a service, new bugs may be introduced, which can eat into an existing error budget. There can then come a stage where the error budget for a given period is exhausted, and management will need to halt development of new features until the error budget is restored. For example, when a company that deploys code changes twice a day consumes half its error budget, it needs to reduce code deployments to once a day. When three-fourths of the error budget is consumed, the deployments can be reduced to once a month. When the error budget hits zero for a month, deployments will need to be halted.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Explain how IT operations need to adapt to thrive in the cloud

References:
[Services.google.com > Increasing Business Value with Better IT Operations: A guide to Site Reliability Engineering \(SRE\)](#)

Related Material

Question #16 of 60

Question ID: 1497150

Your company has various teams and departments, all of which concurrently use resources on a Google Cloud deployment. How should you implement financial governance on Google Cloud to allow your company to achieve more accountability in its cloud operations?

- A) Ensuring all resources are owned by a single user account

✖
- B) Keeping a flat hierarchy with a single organizational level

✖
- C) Managing inventory for on-premises systems

✖
- D) Using cloud resource labels and hierarchies

✔

Explanation

To achieve greater accountability in your cloud deployment for financial governance, you should use cloud resource labels and hierarchies. Using resource hierarchies simplifies the allocation of costs to specific departments and ensures accountability. This action also provides insights into the return on investment (ROI) for your cloud deployment. A resource hierarchy enables you to create a structure for your Cloud account based on your specific business needs. The hierarchy can be described below:

- Level 1 (topmost) – Organization: The Company
- Level 2 – Folders: Departments, shared infrastructure, teams within departments, and products within teams
- Level 3 – Projects: Development, testing, and production using Google Cloud Platform (GCP) for projects
- Level 4 – Resources: Compute Engine instances, Cloud Storage buckets, and App Engine services

Additionally, resource labels can be used to group several resources together and to identify the teams that own particular resources. As an example, resource labels can be App:UI, Env:Prod, and Dept:Testing, which indicate that a resource is for an app called UI in the production environment used by the testing department.

A company needs various financial governance tools for easy cost management that ensures clarity, accountability, and control. Google Cloud financial governance helps a company control costs by providing various tools and services.

- Clarity – This is achieved using billing reports from the Google Cloud Platform (GCP) console, billing application programming interfaces (APIs), and intelligent recommendations.
- Accountability – This is achieved using resource hierarchies and resource labels.
- Control – This is achieved using organizational policies and cost management controls.

Managing inventory for on-premises systems is not a way of implementing accountability on a cloud deployment. An on-premises system and its inventory are independent of cloud resources. The financial management of on-premises systems is not related to the financial governance systems that need to be implemented to manage costs on a cloud deployment.

Ensuring a single user account owns all resources is not a way of implementing accountability on a cloud deployment because a cloud deployment for a company can have several different departments using the system. For this, there needs to be a way to have transparency so the company knows the exact users and departments that utilize specific cloud resources.

Keeping a flat hierarchy with a single organizational level is not a way of implementing accountability on cloud deployment. Keeping a multi-level hierarchy that reflects the operating structure of an organization is critical for providing accurate financial accountability in the cloud deployment. By having various levels of scale, costs can be allocated correctly and managed efficiently based on the specific departments and projects that utilize the resources. With a flat organizational hierarchy, it would be impossible to have accountability for resource utilization and the associated costs of specific departments and projects.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > A guide to financial governance in the cloud](#)

Related Material

Question #17 of 60

Question ID: 1497123

Your company is a gaming application development company. It is considering various cloud service models. Which of these would you be able to do with an IaaS cloud deployment?

- A) Updating the backend infrastructure for virtual machines

✖
- B) Developing code without managing the software development platform

✖
- C) Running testing and development operations

✔
- D) Using an entire application stack from the CSP

✖
- E) Performing analysis for big data

✔

Explanation

Using an Infrastructure as a Service (IaaS) model you can run testing and development operations and perform analysis for big data. IaaS is a cloud service model that provides infrastructure resources to users on demand via the Internet. These resources provide computing, storage, virtualization, and networking services and offer scalability. IaaS provides networking and computing resources, which you can use as a suitable environment for running cycles for testing and development. Also, the scalability of IaaS allows you to combine various disaster recovery (DR) systems into a single virtualized system. IaaS can be used for big data analysis as it has the processing capabilities for handling massive workloads.

You would not use an entire application stack from the cloud service provider (CSP) when using IaaS. This is because it is a feature of the Software as a Service (SaaS) model and not of IaaS. A SaaS model provides full applications and the underlying infrastructure, which can be used by a customer through the Internet.

You would not develop code without managing the software development platform on the IaaS model. This is a feature of the Platform as a Service (PaaS) model where the CSP provides all the necessary software and tools you need to create applications and also maintain the software development platform.

You would not update the backend infrastructure for virtual machines when using an IaaS model. This is the responsibility of the CSP and not the customer in an IaaS model. The customer is responsible for managing the OS, data, and applications.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why cloud technology is revolutionizing business

References:
[Cloud.google.com > What is IaaS?](#)

Related Material

Question #18 of 60

Question ID: 1497157

Your company's accounting department has raised an issue regarding the high operating costs of the cloud deployment. You are asked to find ways to optimize the costs of running the cloud. What should you do to reduce the Total Cost of Ownership (TCO) using Google Cloud?

- A) Store data on the most available storage classes

B) Utilize historical usage data for monitoring

C) Delete idle VM instances

D) Use larger VM sizes than needed
- ✖

✖

✔

✖

Explanation

You should reduce the Total Cost of Ownership (TCO) through cost optimization tactics, such as deleting idle virtual machine (VM) instances. Cost optimization involves using managed services, keeping a consumption model, and analyzing expenditure and attributing it correctly. Cloud monitoring dashboards display key system metrics graphically in real-time, offering an updated view of the cloud system. This helps you get a clear analysis of how your systems infrastructure, applications, and services are doing. This way, you can use metrics and logs to track and resolve issues swiftly, thus improving your TCO and reducing Mean Time to Recovery (MTTR). You can also create alarms based on metrics. An example is ensuring that server sizes are selected to match actual computing needs in order to ensure greater cost-effectiveness.

You would not reduce the TCO by using larger VM sizes than needed. Using larger server sizes than required will cause a waste of resources and thus a waste of capital. Instead, you can opt for rightsizing VM instances based on your specific requirements. For this, you can create a custom machine type that allows you to specify the exact CPU and RAM configuration for your specific processing needs.

You would not reduce the TCO by using historical usage data for monitoring. Historical data cannot give you an updated view of a cloud system. Monitoring involves observing important system metrics in real-time using dashboards. This allows you to find ways to optimize costs through more efficient use of cloud resources. An example is monitoring VM instances to identify resources that may be idle.

You would not reduce the TCO by storing data on the most available storage classes. Using the most available storage class is suitable for high availability applications. However, it is also the most expensive, and not all types of data require high availability. By using more suitable storage classes for specific types of data, you can optimize storage costs and save on TCO. Google Cloud Storage offers various storage classes, which include standard, nearline, coldline, and archival. By moving data to less expensive storage classes, such as the archival class, you can save on costs for storing data that is accessed rarely. By using a lifecycle policy, you can automatically ensure that data objects change their storage class based on conditions you specify, which can be the age of the object.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > Best practices for optimizing your cloud costs](#)

Related Material

Question #19 of 60

Question ID: 1495195

A company has an on-premises PostgreSQL server database system and is looking for a cloud-based alternative that is fully managed so the company can remove overhead costs associated with replication, maintenance, and updates. Which of these solutions should you recommend to meet this requirement?

- A) Cloud SQL

B) Dataproc

C) Cloud Trace

D) Cloud Data Fusion
- ✔

✖

✖

✖

Explanation

You should recommend Cloud SQL for this scenario. Cloud SQL provides a relational database system that is fully managed and works with SQL Server, PostgreSQL, and MySQL. It provides features like flags for configurations, rich extension collections, and ecosystems for developers. Cloud SQL brings down the maintenance costs of databases and provides reliable operations supported by a site reliability engineering (SRE) team that is available around the clock. It has automation built-in for tasks like managing storage capacity and provisioning for databases. It can also be integrated with Google Kubernetes Engine (GKE) and BigQuery while also providing observability of your data using Cloud SQL Insights.

You would not recommend Cloud Trace in this scenario because it is used as a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

You would not recommend Cloud Data Fusion in this scenario because it is a fully managed service that provides scalable integration for cloud-native data. It allows users to deploy Extract, Transform, and Load/Extract, Load, and Transform (ETL/ELT) data without requiring coding by using a visual interface. Cloud Data Fusion is created using the Centralized Data Analysis Platform (CDAP) data analytics platform which is open source.

You would not recommend Dataproc in this scenario because it is better suited for applications related to data lakes. It provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, perform ETL operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > Cloud SQL](#)
[Cloud.google.com > What is Cloud SQL?](#)

Related Material

Question #20 of 60

Question ID: 1495222

The company you work for wants to find more secure and cost-effective methods of managing its systems. You are looking to select the appropriate cloud deployment for this. Which of these deployment configurations uses a hybrid cloud infrastructure?

- A)** Using one public cloud service provider for a web frontend and a different one for the backend
- B)** Using SaaS from one cloud provider and PaaS from another
- C) Using an on-premises data center for storing raw data along with a public cloud for data analytics**
- D)** Storing all organizational data and applications in an on-premises data center

Explanation

Using an on-premises data center for storing raw data along with a public cloud for data analytics is an example of a hybrid cloud infrastructure. In a hybrid cloud system, a company uses its own data center along with a public cloud system to achieve its business goals. A hybrid cloud also always contains a private cloud. A hybrid cloud deployment allows a company to run its applications on-premises and use the public cloud for bursting when user demands go beyond the limits of the data center.

Using one public cloud service provider (CSP) for a web frontend and a different one for the backend is not an example of a hybrid cloud deployment but of a multi-cloud deployment. A multi-cloud deployment involves using multiple public CSPs where each public cloud services a different function for the company. A company can then use one public cloud for database services and another one for analytics. This lets a company use its preferred cloud provider for specific services. Remember that using multiple public cloud service providers indicates a multi-cloud infrastructure.

Using Software as a Service (SaaS) from one cloud provider and Platform as a Service (PaaS) from another is not an example of a hybrid cloud infrastructure but of a multi-cloud infrastructure. This is because both SaaS and PaaS are public cloud offerings, and using several public clouds is a feature of multi-cloud deployments. The PaaS model allows consumers to install their own applications onto a cloud platform. The provider supplies the necessary operating system and physical hardware. This allows consumers to quickly deploy their own applications without the burden of first acquiring the necessary physical resources and operating system(s) to make it happen. The SaaS model covers the use of software applications by consumers who are running on a provider's cloud infrastructure. These applications can be accessed using thin client interfaces, including mobile devices and web browsers. An example of this is email accessible from a web browser, such as Gmail. The advantage of this model is that a consumer can utilize the application without the need to physically install the application or be concerned about the infrastructure needed to run it.

Storing all organizational data and applications in an on-premises data center is not an example of a hybrid cloud infrastructure but of a private cloud infrastructure. In this infrastructure, all the infrastructure and applications that are part of the cloud or virtualization system are running on systems that are inside a company's on-premise data center.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Hybrid and multi-cloud architecture patterns](#)

Related Material

Question #21 of 60

Question ID: 1497405

You want to ensure maximum security of your sensitive data and also leverage cloud-native services like serverless computing. Which cloud deployment should you opt for?

- A)** Multi-Cloud
- B)** Private Cloud
- C) Hybrid Cloud**
- D)** Public Cloud

Explanation

You should opt for a hybrid cloud infrastructure model. In a hybrid cloud system, a company uses its own data center along with a public cloud system to achieve its business goals. A hybrid cloud always contains a private cloud. A hybrid cloud deployment allows a company to run its applications on-premises and use the public cloud for bursting when user demands go beyond the limits of the data center.

You would not opt for a private cloud infrastructure. In this deployment, all the infrastructure and applications that are part of the cloud or virtualization system are running on systems that are inside a company's on-premises data center. This deployment option does not include systems that are a part of a public cloud service offering.

You would not opt for a public cloud infrastructure. In this deployment, the infrastructure includes applications that are fully functional in a cloud and have no on-premises portions. This deployment option may not be suitable for situations where sensitive data needs to be stored locally and not on a public medium.

You would not opt for a multi-cloud infrastructure. In this deployment, the data is still stored on a public system, which can cause security concerns for the company. A multi-cloud deployment involves using multiple public CSPs where each public cloud provides a different service for the company. This allows a company to use one public cloud for database services and another one for analytics. This deployment option lets a company use a preferred cloud provider for specific services. Remember that using multiple public CSPs indicates a multi-cloud infrastructure.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Hybrid and multi-cloud architecture patterns](#)

Related Material

Question #22 of 60

Question ID: 1495277

To ensure the security of your cloud deployment, senior management has decided to implement the principle of least privilege for specifying user access permissions across the cloud. What will be the impact of using a resource access model based on the principle of least privilege for the security of the cloud?

- A) Cloud resource access needs to be approved by the cloud service provider

✖
- B) A user only has access to the cloud resources necessary for task completion

✔
- C) Users need authorization to use on-premises applications

✖
- D) Only senior management can access resources on the cloud

✖

Explanation

The impact of the least privilege principle in access management is that a user only has access to the cloud resources necessary for task completion. This ensures that users cannot access resources that are outside their scope of work, thus increasing privacy and security on the cloud. This access model ensures that users with malicious intent cannot cause damage to resources outside of their privileges.

The least privilege principle does not imply that only senior management can access resources on the cloud. An access management system that only allows senior management access to cloud resources would not allow for normal use of a cloud deployment and would not provide the company with the growth it seeks from using the cloud. By the principle of least privilege, all employees, regardless of their designation, can have access that is limited by their specific role within the organization.

The least privilege principle does not imply that users need authorization to use on-premises applications. The access in systems in place for an on-premises data center is not related to the Identity and Access Management (IAM) that are used on a cloud deployment for restricting access. The purpose of the IAM systems on a cloud is to keep a unified and secure system for authorizing user access to cloud resources.

The least privilege principle does not imply that cloud resource access needs to be approved by the cloud service provider (CSP). Specifying the access rights to cloud resources for users is not the responsibility of the CSP, like Google, but of the company that is using the cloud. As per the shared responsibility model, the CSP is responsible for the security of the cloud, and the customer is responsible for security in the cloud, which includes managing access rights for users.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe a cloud security approach and Google Cloud security benefits

References:
[Services.google.com > Google Cloud security foundations guide](#)

Related Material

Question #23 of 60

Question ID: 1495255

Your company offers data analytics services using resources on its on-premises systems as well as its cloud deployment. It is now looking for a way to host its APIs both on-premises and on Google Cloud. Which Google Cloud service should you recommend for this?

- A) Cloud Backup and DR

✖
- B) Dataproc

✖
- C) Cloud Trace

✖
- D) Apigee hybrid

✔

Explanation

You should recommend Apigee hybrid for this scenario. With Apigee hybrid, you can place gateways near the source of application programming interface (API) traffic to use existing governance and security infrastructure while also benefiting from cloud-native services for monitoring and analytics. This is made possible by managing and controlling the runtime environment using Apigee hybrid. Apigee is used to build, secure, scale, and analyze APIs at any location with support for the complete API lifecycle. It provides strong systems of API control and visibility. With Apigee, you can increase the value of your digital assets by monetizing your API-based products.

You would not use Cloud Backup and Disaster Recovery (DR) in this scenario. Cloud Backup and DR is a service that provides a way to backup workloads from both Google Cloud and on-premises systems for data protection that is consistent as well as centralized. It allows you to have a single system to protect file systems, database instances, and virtual machines (VMs) for both VMware and Compute Engine. Using application-consistent backups that are incremental-forever, Cloud Backup and DR ensures low values for both the recovery point objective (RPO) and recovery time objective (RTO) of a company, ensuring swift recovery of business operations using backups.

You would not use Dataproc for this scenario. Dataproc provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, Extract, Transform, and Load (ETL) operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not use Cloud Trace in this scenario. Cloud Trace is a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Cloud.google.com > Apigee API Management](#)

Related Material

Question #24 of 60Question ID: 1495191

Your company has various multi-cloud and hybrid cloud environments that store its data. You need to find a way to populate this data lake with data from all of these sources. Which service should you use to perform data ingestion for the data lake?

- A) Cloud Data Fusion

B) Dataproc

C) Cloud Trace

D) Cloud Storage
- ✔

✖

✖

✖

Explanation

You should use Cloud Data Fusion for the process of data ingestion into the data lake. Cloud Data Fusion is a fully managed service that provides scalable integration for cloud-native data. It allows users to deploy Extract, Transform, Load/Extract, Load, Transform (ETL/ELT) data without requiring coding by using a visual interface. Cloud Data Fusion is created using the Centralized Data Analysis Platform (CDAP) data analytics platform, which is open source. This allows for the portability of data pipelines and integration for multi-cloud and hybrid cloud deployments that enable users to remove existing solos and access previously inaccessible data. Additionally, Cloud Data Fusion provides several preset transformations that can be used for stream and batch processing tasks. This boosts collaboration and productivity by allowing connections and transformations to be shared among users.

You would not use Cloud Storage for data ingestion. This is because it is an object storage service and not a data ingestion service. Cloud Storage can be used for Object Lifecycle Management (OLM), where data can be configured so that data is automatically moved to lower-cost storage classes when the data meets specified criteria like a period having passed or a new version of the data created. Cloud Storage also provides data redundancy through various locations for storage buckets. This provides fast response times and disaster recovery (DR) options.

You would not use Dataproc for data ingestion. Dataproc provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, ETL operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not use Cloud Trace for data ingestion. Cloud Trace is used as a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > Cloud Data Fusion](#)

Related Material

Question #25 of 60Question ID: 1495171

Which of these is a key aspect of an IaaS cloud deployment for a customer transitioning from a traditional on-premises data center to accommodate its growing business workloads?

- A) Meeting growing user traffic through dynamic provisioning of virtual computing resources

B) Paying for all resources, including unused resources

C) Taking full responsibility for security of the entire cloud deployment

D) Finding all single points of failure in the IaaS infrastructure
- ✔

✖

✖

✖

Explanation

Using an Infrastructure as a Service (IaaS) cloud deployment versus a traditional on-premises data center removes the need to purchase new IT infrastructure to meet growing user traffic. A key aspect of an IaaS cloud deployment is the ability to meet growing user traffic through dynamic provisioning of virtual computing resources without having to purchase the physical hardware yourself. An IaaS model allows you to access various infrastructural resources as individual services through a cloud service provider (CSP). These resources include storage, networking systems, servers, and virtual machines. This greatly reduces costs and overhead associated with creating and running physical infrastructure using the traditional on-premises data center approach. The traditional on-premises technology-based approach involves organizations purchasing IT resources for their applications and data through a long procurement process. Apart from the resources, they also need the physical space to house the resources as well as the necessary cooling and power systems the infrastructure will require. An organization then also needs to have the necessary trained IT staff to manage and run the infrastructure. This entire investment is not feasible to scale when the business experiences a spike in usage. A usage spike can lead to loss of computing capacity, or it can lead to overprovisioning and unnecessary expenses due to the unutilized resources.

Taking full responsibility for the security of the entire cloud deployment is not an aspect of an IaaS model because the CSP is responsible for the security and maintenance of the cloud infrastructure, which includes networking systems and virtual machines. You, as the customer, are responsible for installing and running the software you need and securing your data that is stored in the cloud.

You will not need to pay for all resources, including unused resources, when using an IaaS model. IaaS resources are accessed using a system called Pay-As-You-Go (PAYG), which ensures that you, as the customer, only pay for the resources you actually use, and you do not pay for any resources that are left unused. IaaS systems also provide monitoring, logging, billing, data resiliency, and security systems to give you greater control over your cloud systems operations.

You will not need to find all single points of failure in the IaaS infrastructure. This is because IaaS cloud infrastructures have fault tolerance and redundancy systems already built into them. Your workloads are run across multiple locations and servers, which removes any single points of failure. The services running on an IaaS system will be available even if individual components of hardware resources experience failure.

Objective:

Introduction to Digital Transformation with Google Cloud

Sub-Objective:

Explain why it is critical for businesses to adopt new technology

References:

[Cloud.google.com > What is IaaS?](#)

Related Material

Question #26 of 60

Question ID: 1497134

You are designing a machine learning (ML) production system for your company. You need to create the ML pipeline in a way that ensures that models meet performance expectations. How would you do this? (Choose two.)

- A) Keep the data validation process at the beginning of the pipeline**

✓
- B) Keep the model validation process at the end of the pipeline**

✓
- C) Perform manual updates to ML models post-deployment**

✗
- D) Keep a one-time training for the ML model pre-deployment**

✗
- E) Maintain a tactical maturity phase for ML applications**

✗

Explanation

Data validation measures need to be placed at the beginning of the pipeline and model validation needs to be placed at the end. Data validation is used to verify the data types and schemas for the fresh data, which is used for retraining an existing machine learning (ML) model. Model validation checks the performance of an ML model's predictions and ensures that the performance is better than its predecessor.

Keeping a one-time training for the ML model pre-deployment will not help meet performance expectations for the ML model. This is because ML models need continuous training (CT). CT helps validate all the components and code of ML systems and checks existing models and schemas. The purpose of CT is to help ML models adapt to changes in data before the models are deployed. This is done automatically. To automate CT reliably, you need to place validation and quality checks for ML training into the existing ML pipeline.

You would not maintain a tactical maturity phase for ML applications as this would not meet the enterprise-level needs of a ML production system. The tactical maturity phase is when a company is only exploring the possibilities of using artificial intelligence (AI) and ML technology. This maturity level is characterized by the use of limited computing resources like data science virtual machines (VMs) for local data processing and training ML models. The tactical maturity level also involves the exploration of AI possibilities through use cases for proof of value (POV) demonstrations. This maturity level sees controlled costs for ML exploration with hardware and compute resources that are dedicated for use by selected data scientists. The exploratory analysis of data is done using ready-for-use AI systems and tools for data wrangling.

You would not perform manual updates to ML models post-deployment. This is not a suitable approach because manual updating can lead to errors and is not a feasible solution in an enterprise that may have a massive number of deployed ML models that need regular updates of their data as well as the implementation of new ML concepts. An automated pipeline-based approach would be ideal for updating ML models. This approach is used in transformational maturity, which involves orchestration and automation of ML and data pipelines end-to-end.

Objective:

Innovating with Data and Google Cloud

Sub-Objective:

Identify Google Cloud's solutions for machine learning and AI

References:

[Services.google.com > Google Cloud's AI Adoption Framework](#)

Related Material

Question #27 of 60

Question ID: 1497137

You want to build greater safety and accountability into your company's AI products. To help you achieve this, you are looking for relevant tools that are part of Google's Responsible AI framework. Which tools would you use? (Choose two.)

- A) Looker**

✗
- B) Cloud Storage**

✗
- C) Cloud Trace**

✗
- D) TensorFlow open-source toolkit**

✓
- E) Model cards**

✓

Explanation

Model cards, TensorFlow open-source toolkit, and Explainable artificial intelligence (AI) are tools used as a part of Google’s Responsible AI framework. Responsible AI tools allow you to inspect and analyze existing AI models to ensure safer products, increased customer trust, and ethically sound innovation. Explainable AI is a collection of frameworks and tools for understanding and interpreting machine learning (ML) model predictions and is integrated with various Google services. Model cards provide a structured way to organize ML Models’ key facts, limitations, and capabilities. TensorFlow is an open-source platform used for ML. TensorFlow can be utilized for integrating the processes required for Responsible AI into an existing workflow for ML.

Cloud Trace is not used as a part of the Responsible AI system, but it is used for tracing purposes. Cloud Trace gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

Cloud Storage is not used as a part of the Responsible AI system. It allows you to store and access data on Google Cloud using a Representational State Transfer (RESTful) service. It also provides features for security and data sharing. Cloud Storage can be used for Object Lifecycle Management (OLM), where data can be configured so that data is automatically moved to lower-cost storage classes when the data meets specified criteria like a period having passed or a new version of the data created. Cloud Storage also provides data redundancy through various locations for storage buckets. This provides fast response times and disaster recovery (DR) options.

Looker is not used as a part of the Responsible AI system. Looker is a platform for business intelligence that allows you to access intelligent data for making better business decisions. It also can be embedded into other applications for providing embedded analytics. Looker can be used to create alerts for proactively sending timely insights related to various business data. It also has dashboards that are interactive which allow for analyzing data in real-time through a trusted data model.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > Responsible AI](#)

Related Material

Question #28 of 60

Question ID: 1495279

Your company offers a data analytics service using Google Cloud that is heavily used by its userbase. The development team wants to add enhancements to the service, while the operations team needs to maintain application availability. How can you increase application reliability and service availability while also bringing in new features to the application?

- A) Keep a common SLO for development and operations teams

B) Change the SLA to meet the developer SLO

C) Allow development and operations teams to define their own SLO

D) Use an on-premises data center for delivering new features
- ✔

✖

✖

✖

Explanation

You should keep a common service level objective (SLO) for development and operations teams. An SLO specifies a definite target for a service, which can be 99.999% uptime. SLOs are defined based on user experience and can address user-centric needs, such as latency. Service level indicators (SLIs) help gauge if SLOs are met. A combination of SLOs and SLIs work towards meeting service level agreements (SLAs), which specify the exact agreement made with the users of the service.

You should not allow development and operations teams to define their own SLO. This is because both teams need to agree on a common SLO. Developers want to add new features, while operations teams want to minimize any possible disruptions to the service and seek to minimize any modifications to the service. This creates a conflict between the teams. Creating a common SLO ensures that there can be an agreement between operations and development teams to meet a common goal of ensuring service uptime while adding new features.

You should not use an on-premises data center for delivering new features. A service that is hosted using Google Cloud will benefit from having it enhanced using cloud-native technologies. Cloud technology and its global infrastructure can help provide availability and reliability that exceeds that of a traditional data center. For example, by using several availability zones, a service can continue to serve customers even if one of the data centers it uses fails. This kind of availability would not be possible using a single on-premises data center.

You should not change the SLA to meet the developer’s SLO. Changing the SLA is not an option because the SLO and SLIs need to work towards meeting the SLA as agreed upon with the users for a particular service. Failure to meet a set SLA can incur financial penalties for the company. SLAs are visible to customers that use a service while SLOs and SLIs are internal to the development and operations team. If you do not meet an SLO, then there is a possibility of not meeting the SLA, which can cause issues for the company.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Explain how IT operations need to adapt to thrive in the cloud

References:
[Services.google.com > Increasing Business Value with Better IT Operations: A guide to Site Reliability Engineering \(SRE\)](#)

Related Material

Question #29 of 60

Question ID: 1495252

Your company uses APIs for providing services to its customers, which span millions across the globe. Now, you want to find a way to specify quotas for your APIs as well as thoroughly monitor their usage. Which Google Service should you use for this?

- A) Firestore

B) Payment Gateway

C) Cloud Endpoints

D) Cloud Functions
- ✖

✖

✔

✖

Explanation

You should use Cloud Endpoints for this. Cloud Endpoints provides a system for managing application programming interfaces (APIs) that enables monitoring, analyzing, and securing APIs as well as specifying quotas for them. You can monitor the key operational metrics for the APIs using Google Cloud Console and gain usage insights with BigQuery, Cloud Trace, and Cloud Logging. You can also utilize Cloud Endpoints Portal to create a website that API users can access to find documentation for the APIs and interact with them. Cloud Endpoints provides API protection using API Keys from Google and JavaScript Object Notation (JSON) Web Tokens. It also has less than 1ms of latency per API call and allows for the deployment of APIs using Google Kubernetes Engine (GKE) and App Engine automatically. Cloud Endpoint utilizes an architecture that is distributed and based on a proxy that uses NGINX (“Engine-Ex”), which is an open source system for load balancing, Hypertext Transfer Protocol (HTTP) caching, and reverse proxy services.

You would not use Payment Gateway for this scenario. With Payment Gateway, you can perform processing of payments that are application programming interface (API)-based. It can scale up to cater to unpredictable surges in usage as well as be extended to third-party insights and anomaly finding tools through APIs. It also allows for payment processing from across multiple regions of Google Cloud’s global infrastructure. Payment Gateway also has built-in tools that allow you to monitor, debug, and trace API-based purchases with high availability and less overhead. You can also export data of transactions into a project on Google Cloud using an export API.

You would not use Firestore for this scenario. Firestore is a serverless document database that is scalable and fully managed. It enables fast development for applications for the internet, mobile, and the Internet of Things (IoT), which can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features.

You would not use Cloud Functions for this scenario. Cloud Functions is best suited for when you need to run functional code based on an event or a trigger. Cloud Functions runs your code when events occur from various sources, such as Pub/Sub, Cloud Storage, and Hypertext Transfer Protocol (HTTP). An example is running a process for video transcoding automatically whenever a video is uploaded to Cloud Storage.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Cloud.google.com > Cloud Endpoints documentation](#)

Related Material

Question #30 of 60

Question ID: 1495940

Your company has multiple teams that have hundreds of employees in them. You need a centralized way to manage user access to cloud resources. Which of these systems would you need to use to authorize which users can access specific Google Cloud resources?

- A) Cloud IAM

B) Secret Manager

C) Billing APIs

D) Encryption at rest
- ✔

✖

✖

✖

Explanation

You would use Cloud Identity and Access Management (IAM) for this scenario. Cloud IAM enables you to manage cloud resources centrally with granular access control and visibility. With IAM, you can grant access at the resource level, such as providing the role of Subscriber to a user for a Pub/Sub topic. This way, IAM allows you to grant roles in addition to Owner, Editor, and Viewer roles. IAM also provides Recommender that can enable you to detect overly permissive access policies and rightsize them. With IAM, resource access is controlled using contextual attributes, such as IP address or resource type. You can perform IAM policy management and creation using the Google Cloud Console, programmatically using IAM methods, or by command-line access using the gcloud tool. Google Accounts are also supported by IAM. You can utilize Cloud Identity with IAM to synchronize user accounts across multiple projects. IAM also provides workforce identity federation that enables you to perform authentication and authorization of a workforce using external identity providers (IdPs). IAM can provide a full audit history that includes granting, removal, and delegation of permissions for compliance needs.

You would not use Secret Manager for this scenario. You use Secret Manager for storing sensitive data like passwords, application programming interface (API) keys, and certificates securely. With Secret Manager, you have a single source of truth for accessing, managing, and auditing secrets on a Google Cloud deployment. Secret Manager provides lifecycle management with versioning for pinning requests to the newest versions of secrets, as well as the ability to use Cloud Functions to automate the process of rotating secrets. It also allows you to select regions for the replication of your secrets data. Secret Manager is integrated with Cloud IAM for controlling access to secrets.

You would not use encryption at rest in this scenario. Encryption at rest of stored data is a way of implementing security for data stored on the cloud infrastructure. Google utilizes multiple layers of encryption for the protection of data at rest. Google's storage infrastructure encrypts all user data by default before it is stored on physical storage. This way, encryption is performed by Google Cloud at the storage or application layer of the cloud infrastructure. Google has several storage services like Colossus and Cloud Spanner, as well as a key management service.

You would not use billing APIs for this scenario. You can use billing APIs to access various tools created by other vendors to manage project bills programmatically and obtain data on all stock keeping units (SKUs) in their repository and related metadata. This is a way of ensuring clarity for financial governance on a cloud deployment. Other methods include using intelligence recommendations for performing rightsizing of computing resources. This helps ensure that costs are minimized by ensuring that virtual machine (VM) usage is optimized.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe a cloud security approach and Google Cloud security benefits

References:
[Cloud.google.com > Cloud Identity and Access Management \(IAM\)](#)

Related Material

Question #31 of 60

Question ID: 1495186

An e-commerce company is expanding its workforce and is working on performing data analysis to find ways to grow its business. You need to use a suitable service to improve decision making and productivity in the organization by providing key insights to a growing number of users. Which tool should you use for this requirement?

- A) Dataproc

B) Cloud Trace

C) Dataflow

D) Looker
- ✖

✖

✖

✔

Explanation

You should use Looker. Looker is a platform for business intelligence (BI) that allows you to access intelligent data for making better business decisions. It can be embedded into other applications for providing embedded analytics. Looker can be used to create alerts for proactively sending timely insights related to various business data. It also has dashboards that are interactive, which allow for analyzing data in real-time through a trusted data model.

You would not use Dataproc for this scenario. Dataproc provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, Extract, Transform, and Load (ETL) operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not use Dataflow for this scenario. Dataflow is a fully managed service that automates the process of provisioning and managing processing resources. It provides the highest possible resource utilization by performing horizontal autoscaling for various worker resources.

You would not use Cloud Trace in this scenario as it is not for BI, but a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > Looker](#)

Related Material

Question #32 of 60

Question ID: 1495941

Your company has various business services in use that require the procurement of third-party software. You need to do a security analysis of this approach. What can be the potential problems that can occur when using third-party software in an existing cloud deployment?

- A) Third-party software lacks the support systems available for cloud-based software

✖
- B) Third-party software reduces a company's innovation goals

✖
- C) Third-party software cannot handle important business services

✖
- D) Third-party software can be prone to cyberattacks

✔

Explanation

Third-party software can be prone to cyberattacks. Third-party software can have security issues in it, which can compromise an organization's data security and make the organization vulnerable to cyberattacks. There needs to be a comprehensive security architecture in place within a company's IT processes to check third-party software and monitor it on a regular basis to ensure there are no security issues.

Third-party software has ample support systems available through various channels that include vendor support, online forums, and support groups. Third-party software can be developed by companies or groups that have a strong grasp of cybersecurity challenges and take the necessary measures and perform development best practices to ensure the software is secure and error free. The security level of software depends greatly on the development methodology employed to create it as well as the level of security awareness of the team that created it. Some open-source frameworks have forums and collaborative groups where open knowledge sharing contributes to mitigating any existing issues.

Third-party software does not reduce a company's innovation goals. The level of innovation that a company can incorporate into its operation depends more on the scale to which it has embraced digital transformation and not on its reliance on third-party software. To reach a state of being fully innovative, an organization will first need to ensure that it has open knowledge sharing, decentralized teams, more autonomous decision making, and use of artificial intelligence (AI) systems by team members of varying technical skill.

Third-party software can handle important business services and serve various critical functions of an organization. The amount of capability available in third-party software varies across industries and vendors. Third-party software can be as powerful as cloud offerings and sometimes even exceed the current capabilities of a cloud software offering. You need to select the right third-party software for the task at hand and make the decision based on various factors, which include the security of the third-party software.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe a cloud security approach and Google Cloud security benefits

References:
[Cloud.google.com > Google security overview](#)

Related Material

Question #33 of 60

Question ID: 1495925

A company is consulting with you to quickly scale their existing system architecture, which utilizes virtual machines. You decide to recommend VMware Engine to them. Which of these will the company be able to achieve with this recommendation?

- A) Perform migration of existing virtual machines to new containers

✖
- B) Utilize a fully managed and highly available file service

✖
- C) Perform replatforming of virtual machines without changing their location

✔
- D) Perform monitoring, verification, and optimization tasks for a network

✖

Explanation

Using VMware Engine, the company can perform replatforming of virtual machines (VMs) without changing their location. With VMware Engine, you can move applications that are based on VMware onto Google Cloud without any changes to existing applications. It provides all the necessary infrastructure and VMware licenses that allow you to have a software-defined data center (SDDC) on Google Cloud. Using VMware Engine provides elasticity and scaling through the ability to swiftly build a private cloud using flexible management of resources and automatic scaling.

You cannot perform migration of existing VMs to new containers using VMware Engine. A key difference between containers and VMs is that containers perform virtualization at the level of the OS while VMs perform virtualization at the level of hardware. Containers are software packages that contain application code as well as dependencies like runtime environments and libraries that are required for running an application. This allows containers to virtualize an OS and allow applications to be made portable across multiple systems.

You cannot perform monitoring, verification, and optimization tasks for a network using VMware Engine. You can use Network Intelligence Center for that. Network Intelligence Center is used for monitoring, verifying, and optimizing a network and can be used for Google Cloud, on-premises, and multi-cloud systems. It provides a reduction in the risks of outages occurring and also ensures greater levels of compliance with regulations and security.

You cannot utilize a fully managed and highly available file service using VMware Engine. For that, you can use Cloud Filestore. You would use Cloud Filestore for enterprise-grade data applications that need storage with durability and persistence, which can be accessed using a Network File System (NFS). You can also use Cloud Filestore for storage systems that need a file system that is Portable Operating System Interface (POSIX) compliant.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Google Cloud VMware Engine](#)

Related Material

Question #34 of 60

Question ID: 1497156

Your company's C-Suite has asked you to take over the task of directing the management of its ongoing Google Cloud expenses. Which of these actions must a company perform to achieve better financial governance of its cloud deployment?

- A) Implement throttling on cloud resources exceeding quotas**

✔
- B) Ensure the on-premises data center matches the cloud for hardware**

✖
- C) Keep teams siloed for better individual focus on their work**

✖
- D) Allow all users across the company to provision and access resources**

✖

Explanation

A company must implement throttling on cloud resources that exceed quotas to achieve better financial governance of a cloud deployment. Setting resource quotas and performing throttling needs to be done to ensure that cost overruns do not occur due to resource utilization that exceeds the specified budgets. A company can keep various controls when considering a public cloud deployment. These can help ensure better governance over cloud expenditure.

- Specify permissions for cloud resource usage using the principle of least privilege
- Maintain control of cloud resource usage by implementing resource quotas
- Control cloud expenditure using budgetary action automation and notifications

You would not allow all users across the company to provision and access resources. This can cause improper use of cloud resources that leads to wasted capital as well as security issues. Instead, you would opt to use the principle of least privilege for specifying access permissions. The impact of the least privilege principle in access management is that a user only has access to the cloud resources necessary for task completion. This ensures that a user cannot access resources that are outside their scope of work, thus increasing privacy and security on the cloud. This access model ensures that users with malicious intent cannot cause damage to resources outside of their privileges.

You would not ensure the on-premises data center matches the cloud for hardware. The on-premises data center is independent of the cloud deployment in its infrastructure. The purpose of using a cloud deployment is to have a cost-optimized way of scaling computing and storage resources to match business demands without needing to take on the capital expenditure associated with procuring the same level of infrastructure required for a data center.

You would not keep teams siloed for better individual focus on their work. By incorporating a partnership that is implemented across various teams, you will gain greater visibility, which is a necessity for efficient cost management. Reducing budget overruns and unauthorized resource use can be accomplished with more collaboration between IT and finance teams.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > A guide to financial governance in the cloud](#)

Related Material

Question #35 of 60

Question ID: 1495283

You work for a software development company that makes two deployments a day for its workplace collaboration tool, which is heavily in use and growing daily. Your development team has come up with a lot of new features that it wants to release, but the specified error budget for the month has been consumed. How will you roll out future deployments?

- A) Once a day**

✖
- B) Once a week**

✖
- C) Once a month**

✖
- D) Halt deployments**

✔

Explanation

You will halt deployments. This is because when new features are added to a service, new bugs may be introduced, which can eat into an existing error budget. There can then come a stage where the error budget for a given period is exhausted, and management will need to halt development of new features until the error budget is restored. For example, when a company that deploys code changes twice a day consumes half its error budget, it needs to reduce code deployments to once a day. When three-fourths of the error budget is consumed, the deployments can be reduced to once a month. When the error budget hits zero for a month, deployments will need to be halted.

The other options are incorrect because you cannot roll out any new deployments when the error budget is consumed for the month. This can cause a failure to meet the service level agreement (SLA) with the service users and cause financial loss for the company. The error budget is the measure of the system's possible downtime, which is the difference between the service level objective (SLO) and the total availability figure of 100%. When an SLO is agreed upon, its corresponding service downtime or error budget is also agreed upon. An SLO specifies a definite target for a service, which can be 99.999% uptime. SLOs are defined based on user experience and can address user-centric needs, such as latency.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Explain how IT operations need to adapt to thrive in the cloud

References:
[Services.google.com > Increasing Business Value with Better IT Operations: A guide to Site Reliability Engineering \(SRE\)](#)

Related Material

Question #36 of 60

Question ID: 1497132

Your company is using an ML model or making sales predictions based on customer shopping preferences. So far, the predictions have not been good. Which of these could be the cause of poor prediction quality in the ML model?

- A) Usage of skewed data**

✔
- B)** Placing data validation at the start of the ML pipeline

✖
- C)** Having transformational maturity in ML applications

✖
- D)** Performing continuous training

✖

Explanation

Skewed and anomalous data are primary cause of bad predictions made by machine learning (ML) models. ML needs to have some quality control measures implemented for the best results. These measures include keeping processes to evaluate models after training and to validate them pre-deployment. Additionally, there needs to be logging and monitoring systems in place so that the system's health can be checked, and notifications can be issued in the event of any anomaly. When continuous integration/continuous delivery (CI/CD) is applied to ML, it involves validation of ML models, data schemas, and code as part of CI, and deployment of ML models and continuous testing (CT) pipelines as part of CD. To create a production system for ML, you would use ML Metadata, Container Registry Cloud Build, and Model Registry services.

Continuous training (CT) is not a cause for bad predictions made by ML models. CT helps validate all the components and code of ML systems and checks existing models and schemas. The purpose of CT is to help ML models adapt to changes in data before the models are deployed. This is done automatically. To automate CT reliably, you must place validation and quality checks for ML training into the existing ML pipeline. Data validation measures need to be placed at the beginning of the pipeline and model validation needs to be placed at the end. Data validation is used to verify the data types and schemas for the fresh data, which is used to retrain an existing ML model. Model validation checks the performance of an ML model's predictions and ensures that the performance is better than its predecessor.

Transformational maturity in ML applications is not a cause for bad predictions made by ML models. Transformational maturity has systems in place to ensure optimum performance of ML models. Transformational maturity is characterized by the creation of systems for the online processing of real-time events and the creation of a unified ML platform utilized by ML specialists across multiple teams. There is also a provision of compute resources, recommended practices, tools, and services needed by ML teams to complete ML projects. This maturity level also sees the use of tools to accelerate ML computing like Cloud Tensor Processing Units (TPUs) and Graphics Processing Units (GPUs) and the use of templates for data processing, which are driven by metadata so workflows can be deployed without the need for coding. Transformational maturity involves orchestration and automation of ML and data pipelines end-to-end.

Placing data validation at the start of the ML pipeline is not a cause for bad predictions made by ML models. Data validation needs to be kept at the start of the pipeline. This is because the data types and schemas used for model training are correct. Without this step, the data fed to an ML model may not be accurate, which leads to poor results in ML model predictions. Additionally, you need to place the model validation process at the end of the pipeline so that you can verify whether or not the trained model is meeting prediction expectations better than the current ML model in use.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Services.google.com > Google Cloud's AI Adoption Framework](#)

Related Material

Question #37 of 60

Question ID: 1497149

A bank has an on-premises data center that it needs to maintain to ensure regulatory compliance. It also has several applications that it uses for internal purposes as well as for serving its customers. What should you use to provide a single portal for visibility of their nodes and clusters in their cloud and on-premises workloads that also manages large volumes of Kubernetes clusters inside their applications?

- A)** Cloud Functions

✖
- B) Anthos**

✔
- C)** App Engine

✖
- D)** Firestore

✖

Explanation

You should use Anthos for this scenario. Using Anthos, you can create and manage an application and then run it on both on-premises systems and on the cloud. With Anthos, you can perform migrations straight from virtual machines (VMs), and create and deploy apps using Google Kubernetes Engine (GKE), VMs, and landing zones that are serverless. Anthos allows you to use its platform for deploying both on-premises and cloud-native applications to create a global fleet that has consistency in its operations. Using containers and serverless services, Anthos lets you grow productivity. By maintaining security systems at every stage of the application’s lifecycle, Anthos provides protection for both a software supply chain and application.

You would not use App Engine for this scenario. App Engine allows you to create and host your applications on the same platforms Google uses for several of its own applications. With App Engine, you can build and deploy applications quickly and manage them easily. It provides scalability without the issues related to backups, patching, and hardware provisioning. App Engine provides support for web application development using programming languages like Python, Ruby, C#, Node.js, Go, and PHP.

You would not use Cloud Functions for this scenario. Cloud Functions is best suited for when you need to run functional code based on an event or a trigger. Cloud Functions runs your code when events occur from various sources, such as Pub/Sub, Cloud Storage, and Hypertext Transfer Protocol (HTTP). For example: running a process for video transcoding automatically whenever a video is uploaded to Cloud Storage.

You would not use Firestore for this scenario. Firestore is a serverless document database that is scalable and fully managed. It enables fast development for applications for the internet, mobile, and the Internet of Things (IoT), which can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features. It also allows you to customize various rules for security and data validation to protect your data. Firestore can be integrated with BigQuery and Firebase. Firestore is an upgrade to the older service called Datastore, which provides a NoSQL database that is scalable.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand modernizing applications with Google Cloud

References:
[Cloud.google.com > Anthos](#)

Related Material

Question #38 of 60

Question ID: 1495297

You need to store the events generated by various Google Cloud services in buckets that need to be localized to specific regions for regulatory and compliance requirements. Which Google Cloud service would be the most appropriate choice for this?

- A) Cloud Trace

✖
- B) Cloud Profiler

✖
- C) Cloud Logging

✔
- D) Cloud Endpoints

✖

Explanation

You would use Cloud Logging for this scenario. Cloud Logging is a log management service that is fully managed. It allows you to store, search, and analyze exabytes of log data as well as set up alerts and notifications. Cloud Logging has Logs Explorer, which lets you use flexible queries for searching, sorting, and analyzing logs. You can also utilize log buckets to store and process log data based on regional compliance requirements. Cloud Logging also has an Error Reporting feature that performs a smart collection of errors in logs into error groups. Cloud Logging also allows your security teams to keep audit trails using its Cloud Audit Logs feature, which provides greater transparency in activities related to administration. Cloud Logging uses the Cloud Logging Application Programming Interface (API) for collecting log entries, which are passed through a Logs Router. The Logs Router uses existing filters to identify the log entries meant for exclusion, ingestion, or import.

You would not use Cloud Profiler for this scenario. Cloud Profiler is used to find and remove any possible performance issues from your applications currently in production. This is accomplished by Cloud Profiler regularly profiling the consumption of resources across your production applications. Cloud Profiler creates an analysis of the performance of an application by collecting data related to CPU usage and memory allocation.

You would not use Cloud Trace for this scenario. Cloud Trace is a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights into its performance.

You would not use Cloud Endpoints for this scenario. Cloud Endpoints provide a system for managing APIs that enables monitoring, analyzing, and securing APIs as well as specifying quotas for them. You can monitor the key operational metrics for the APIs using Google Cloud Console and gain usage insights with BigQuery, Cloud Trace, and Cloud Logging.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Identify Google Cloud solutions for cloud resource monitoring and application performance management

References:
[Cloud.google.com > Cloud Logging](#)

Related Material

Question #39 of 60

Question ID: 1495242

Your company uses a hybrid cloud architecture with several of its applications needing to run on its on-premises data center as well as on Google Cloud. You need to recommend a service that can meet this requirement. Which service should you recommend?

- A) App Engine

✖
- B) Compute Engine

✖
- C) Anthos

✔
- D) Firestore

✖

Explanation

You should recommend Anthos. Using Anthos, you can create and manage an application and then run it on both on-premises systems and on the cloud. With Anthos, you can perform migrations straight from virtual machines (VMs), create and deploy apps using Google Kubernetes Engine (GKE), VMs, as well as landing zones that are serverless. Anthos allows you to use its platform to deploy both on-premises and cloud-native applications to create a global fleet that has consistency in its operations. Using containers and serverless services, Anthos lets you grow productivity. By maintaining security systems at every stage of the application's lifecycle, Anthos provides protection for both a software supply chain and an application.

You would not recommend App Engine. This cannot be used on-premises because it is a cloud computing platform. App Engine allows you to create and host your applications on the same platforms Google uses for several of its own applications. With App Engine you can build and deploy applications quickly and manage them easily. It provides scalability without the issues related to backups, patching, and hardware provisioning.

You would not recommend Compute Engine. This cannot be used on-premise because it is Infrastructure as a Service (IaaS). Compute Engine provides VM computing that has scalability and flexibility. It allows you to utilize Cloud Tensor Processing Units (TPUs), Graphics Processing Units (GPUs), and CPUs to perform analytics and processing on a large scale using Google's technological infrastructure.

You would not recommend Firestore. Firestore is a serverless document database which is scalable and fully managed. It enables fast development for applications for the internet, mobile, and Internet of Things (IoT), which can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features. It also allows you to customize various rules for security and data validation to protect your data. Firestore can be integrated with BigQuery and Firebase. Firestore is an upgrade to the older service called Datastore, which provides a NoSQL database which is scalable.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand modernizing applications with Google Cloud

References:
[Cloud.google.com > Anthos](#)

Related Material

Question #40 of 60

Question ID: 1495229

You need to find the right cloud infrastructure option for migrating from a legacy system with specific licensing needs and running your workloads. Which Google service should you use when you need to have granular control over infrastructural resources, such as CPU and RAM, on a particular OS?

- A) Cloud Functions**
- B) Cloud Run**
- C) App Engine**
- D) Compute Engine**
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Explanation

You should use Compute Engine for this scenario. You would use Compute Engine when migrating from a legacy system that has specific OS and licensing needs. Compute Engine provides the use of virtual machines (VMs) where you can specify the exact memory, CPU, disk storage, and graphics processing units (GPUs) you need, along with the exact OS and other applications you need to use.

You would not opt for App Engine in this scenario. App Engine provides a serverless platform that is fully managed for building and hosting full web applications. You can write a web application using a preferred language and then use App Engine to manage the tasks related to application and database scaling as well as performing updates to the application. App Engine is the preferred option for when you need to focus on coding a web application or a backend for a mobile app and leave the tasks of deployment and scaling to Google Cloud.

You would not opt for Cloud Functions in this scenario. Cloud Functions is best suited for when you need to run functional code based on an event or a trigger. Cloud Functions runs your code when events occur from various sources, such as Pub/Sub, Cloud Storage, and Hypertext Transfer Protocol (HTTP). For example: running a process for video transcoding automatically whenever a video is uploaded to Cloud Storage.

You would not opt for Cloud Run in this scenario. You can use Cloud Run for situations where you need to perform deployment of containerized applications that are written in your preferred coding language. Cloud Run provides support for HTTP, Hypertext Transfer Protocol Secure (HTTPS), and websockets. You can use Cloud Run for creating application programming interfaces (APIs), applications for processing data, webhooks, and websites. With Cloud Run, code can be packaged into a container and simply deployed without needing infrastructural scaling. Another related container technology is Google Kubernetes Engine (GKE), which allows you to create and manage Kubernetes clusters. Kubernetes provides an open-source system to automatically manage, deploy, and scale container-based applications. GKE provides detailed control over containers and nodes, enabling you to specify CPU, RAM, storage, GPUs, and network configurations, along with an Autopilot option for situations where there is not enough operational and developer support available.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Where should I run my stuff? Choosing a Google Cloud compute option](#)

Related Material

Question #41 of 60

Question ID: 1495227

You want to create a cloud-based testing environment that duplicates various different hardware setups so that the testing team can perform various application tests within this isolated environment. Which of these technologies should you use for this requirement?

- A) Containers**
- B) Serverless data warehouses**
- C) Virtual machines**
- D) Physical computers**
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Explanation

This is a use case for virtual machines (VMs). A VM is a virtual computer that runs operating systems and applications. Multiple VMs can run on a single physical computer and access various resources from a hypervisor. A hypervisor is software that creates an abstraction of all the computing resources of the physical computer and provides these resources to VMs running on the computer as required. The VM architecture provides the underlying framework in use for modern cloud computing. Compute Engine is Google's VM service, which offers VMs that can be customized. Some uses of VMs in enterprises include:

- Server consolidation – VMs can act as servers to host other VMs and so reduce sprawl by consolidating multiple computing systems into a single physical machine.
- Migration of workloads – VMs are flexible and portable, which enable swift workload migration for companies.
- Test environment creation – VMs can provide an isolated system that can be used for testing without affecting the surrounding systems.
- Disaster recovery – You can duplicate an existing system as a VM on the cloud, thus adding an additional layer of security to the system.
- DevOps – VMs can be activated, deactivated, and migrated smoothly, which facilitates flexible app development.
- Hybrid systems – VMs help you flexibly integrate existing on-premises legacy systems with a cloud environment to create a hybrid computing environment.

You would not use containers for this scenario. Containers are better suited for creating applications that need to be compartmentalized and made portable across multiple platforms. By compartmentalizing an application using containers, you can then change parts of the application in isolation from the other parts. Containers are software packages that contain application code as well as dependencies, such as runtime environments and libraries, that are required for running an application. This allows containers to virtualize an OS and allow applications to be made portable across multiple systems.

You would not use serverless data warehouses for this scenario. A serverless system like BigQuery would be suitable for a requirement where there needs to be very high levels of scalability for working with very large quantities of data. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and that can be utilized for analytics. It has features like machine learning (ML), business intelligence (BI), and geospatial analytics ready for use. It also has a serverless architecture that allows you to use SQL queries to gain key organizational data. BigQuery has a scalable analysis engine that has a distributed structure and allows you to perform queries on petabytes of data within minutes.

You would not use physical computers for this scenario. This is because procuring physical computers for creating testing environments is not as economical, feasible, or flexible as using a VM based system. A single physical computer has one hardware configuration with a specific CPU, RAM, and storage setup. Using several different VMs, you can have several hardware setups, which makes the process of testing an application across multiple platforms much simpler.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > What is a Virtual Machine?](#)

Related Material

Question #42 of 60

Question ID: 1495245

Your company is looking for a serverless platform that is fully managed for running containers individually. What is the most efficient way to write code for an application and deploy it as a container with no additional overhead required?

- A) Cloud Run**
- B) Cloud Functions**
- C) Compute Engine**
- D) App Engine**
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Explanation

You should use Cloud Run for this scenario. You can use Cloud Run for situations where you need to perform deployment of containerized applications that are written in your preferred coding language. Cloud Run provides support for Hypertext Transfer Protocol (HTTP), Hypertext Transfer Protocol Secure (HTTPS), and websockets. You can use Cloud Run for creating application programming interfaces (APIs), applications for processing data, webhooks, and websites. With Cloud Run, code can be packaged into a container and simply deployed without needing infrastructural scaling. The apps containerized with Cloud Run can be written in Ruby, Python, Go, Java, Node.js, or .NET.

You would not use Cloud Functions for this scenario. Cloud Functions is best suited for when you need to run functional code based on an event or a trigger. Cloud Functions runs your code when events occur from various sources like Pub/Sub, Cloud Storage, and Hypertext Transfer Protocol (HTTP). An example is running a process for video transcoding automatically whenever a video is uploaded to Cloud Storage.

You would not use Compute Engine for this scenario. Compute Engine provides virtual machine (VM) computing that has scalability and flexibility. It allows you to utilize Cloud Tensor Processing Units (TPUs), Graphics Processing Units (GPUs), and CPUs to perform analytics and processing on a large scale using Google's technological infrastructure.

You would not use App Engine for this scenario. App Engine allows you to create and host your applications on the same platforms Google uses for several of its own applications. With App Engine, you can build and deploy applications quickly and manage them easily. It provides scalability without the issues related to backups, patching, and hardware provisioning. App Engine provides support for web application development using programming languages like Python, Ruby, C#, Node.js, Go, and PHP.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand modernizing applications with Google Cloud

References:
[Cloud.google.com > Cloud Run](#)

Related Material

Question #43 of 60

Question ID: 1497161

A company has a sketching app used by millions of users around the world, and now it needs to make some changes to the application's front end. Which would be the recommended course of action as per Site Reliability Engineering (SRE) best practices?

- A) Release the modification to all users to ensure coverage**
- B) Release the modification to a small subset of users for testing**
- C) Do not make the release until there is a zero chance of errors**
- D) Do not make the release until automation for toil is removed**
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Explanation

As per Site Reliability Engineering (SRE) best practices, it is recommended that the company release the modification to a small subset of users for testing. A key SRE best practice is ensuring that changes are released gradually with testing performed for the changes. Some SRE best practices for production servers include:

- Control failures – Ensure failures don't impact the functionality and allow systems to work using a known configuration and only use new data or configurations after they have been sanitized and validated by a human user.
- Make rollouts progressive – All rollouts and updates that need to be performed should be performed in stages and monitored throughout. The changes should be applied to a small section of traffic across time frames and geographies. This allows for problems to be detected and mitigated.
- Specify service level objectives (SLOs) from a user’s perspective – This involves setting measures for performance and availability using metrics that are applicable to a user. An example of this is the latency and error rates of a client application.
- Maintain a balance between error budgets and innovation – The system’s reliability and speed of performing innovation need to be in balance with error budgets, which are the acceptable failure levels of a specified service. An example is a 0.01% error budget for a service with 99.99% availability. This implies that a team can plan to add new features and innovation to a service only if the service has not already availed its error budget for the month, which includes errors and downtime.
- Performing monitoring – System monitoring for health can yield outputs that include pages, tickets, and logging, and require human intervention depending on the severity of a reported issue.
- Do blameless postmortems – This ensures that systems are improved in their design so that complete groups of possible errors that could arise are mitigated. It is done by not blaming people and focusing on making changes to technology and processes.
- Plan for capacity – This includes making plans to handle planned and unplanned outages while ensuring it does not affect a user beyond an acceptable limit. It involves performing load testing to find resource-to-capacity relationships and validating demand forecasts with actual demands until the two match consistently.
- Checking for overloads – This involves making sure that services still produce acceptable results even if overloaded so that users can continue using them. It may require certain features of a service to be halted in an overload situation, while the main service features are still made available.
- Limit operational tasks for SRE teams – This is to make sure that SRE teams do not dedicate more than half of their time to performing operational tasks (toil) that should be sent to development teams. For this, you can ensure there are regular meetings between SRE and development teams. Toil comprises work that may be manual, nonenduring, automatable, repetitive, and tactical.

You must not release the modification to all users to ensure coverage. As per SRE best practices, until a rollout or modification is tested and verified by a small set of users across various geographies and time frames, it must not be made available to the full set of users. This is to ensure any bugs or errors are mitigated before the change reaches all users,

Not making the release until there is a zero chance of errors is not a part of SRE best practices. Errors and failures are a natural part of the SRE process, but failure needs to be dealt with in a controlled manner. As per SRE best practices, you need to ensure that a steady state of the service is maintained so that any failure that occurs is contained and does not cause a catastrophic failure of the entire service.

Not making the release until automation for toil is removed is not an SRE best practice. On the contrary, a key goal of SRE is to ensure toil automation takes place. Toil automation removes the need for performing manual, repetitive work, which does not have an enduring value.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Explain how IT operations need to adapt to thrive in the cloud

References:
[SRE: A Collection of Best Practices for Production Services](#)

Related Material

Question #44 of 60

Question ID: 1497153

You are a consultant for a stock trading company that is setting up its Google Cloud environment. You need to guide the team on some key issues that the company needs to be aware of when managing cloud environments. Which of these concerns should you specify? (Choose two.)

- A)** Security of servers housing virtual machines
- B)** More overhead using cloud services
- C)** Lack of on-premises servers
- D) Lack of cost predictability**
- E) Less cloud usage transparency**

Explanation

You should specify lack of cost predictability and less cloud usage transparency as key issues the company should be aware of. Lack of cost predictability can be an issue for companies managing cloud environments. A customer application may experience surges in user traffic alternated with periods of low levels of traffic. Due to the unpredictable nature of application usage, it can become difficult for a cloud administrator to specify resource limits. Setting resource limits too low can lead to application performance issues during peak usage, and provisioning more resources than required can lead to wasted capital. Less visibility and transparency in cloud usage leads to the wastage of cloud expenditure due to unutilized resources. By using cloud monitoring tools, a cloud consumer can identify resources that are unused and deprovision them so that they do not contribute to the overall billing. This helps reduce unnecessary costs for a cloud deployment.

The lack of on-premises servers is not a concern when managing a cloud environment because the set-up of an on-premises data center and its operation are not related to a cloud environment. To increase the computing power of a cloud deployment, you can perform scaling either manually or automatically, which provisions more virtual servers when required.

More overhead using cloud services is not a concern when managing a cloud environment. By using fully managed services on Google Cloud, you can reduce your operating costs by removing all the overhead that would be required for various tasks, which include resource provisioning, backups, updates, and patching. This reduction in costs allows you to better utilize your capital to grow your business.

The security of servers housing virtual machines (VMs) is not a concern when managing a cloud environment. This is the responsibility of the cloud service provider (CSP). As per the shared responsibility model of cloud environments, the customer is responsible for securing the data and applications that are hosted on the cloud, and the CSP secures the entire cloud infrastructure, which includes networking systems and physical servers that house VMs.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com >A guide to financial governance in the cloud](#)

Related Material

Question #45 of 60

Question ID: 1497125

You work for a data sciences company that is exploring cloud-based services for modernizing its data processing systems. You need to find a secure, fully managed service for managing clusters with Google Compute and Kubernetes systems. Which service will you use?

- A)** Dataflow
- B) Dataproc**
- C)** BigQuery
- D)** Dataprep

Explanation

You will use Dataproc for this scenario. Dataproc provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, Extract, Transform, and Load (ETL) operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not use BigQuery for this scenario. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and that can be utilized for analytics. It has features like machine learning (ML), business intelligence, and geospatial analytics ready for use. It has a serverless architecture that allows you to use SQL queries to gain key organizational data. BigQuery has a scalable analysis engine that has a distributed structure. This enables you to perform queries on petabytes of data within minutes.

You would not use Dataprep for this scenario. Dataprep is a data service for exploring, cleaning, and preparing data visually for the purpose of ML and analysis. It is an intelligent data service and is serverless. This removes the need for users to deploy or manage infrastructure. It also suggests data transformations that are ideal each time a user interface (UI) input is detected, and this removes the need for a user to write code for these applications. Dataprep also detects any anomalies that may exist in the data, including duplicates and missing value data. This helps a user avoid having to assess the quality of data and proceed directly to exploratory data analysis.

You would not use Dataflow for this scenario. Dataflow is a fully managed service that automates the process of provisioning and managing processing resources. It provides the highest possible resource utilization by performing horizontal autoscaling for various worker resources.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for data management

References:
[Cloud.google.com > Dataproc](#)

Related Material

Question #46 of 60

Question ID: 1497158

Your company is a data processing company that has teams around the world. You need to optimize the costs of your cloud deployment while making datasets available globally for faster access for teams. Which of these cost management strategies should you use for the business?

A) Configure a lifecycle policy for data objects

B) Use multi-region buckets for data storage

C) Create multiple copies of data in several buckets

D) Set auto stop for VM instances

Explanation

You should use multi-region buckets for data storage. By using multi-region buckets, you can provide greater resiliency to teams distributed globally. This also removes the need for creating duplicate data, which is not a cost optimized approach. Multi-region buckets provide asynchronous replication across multiple geographic locations and also provide backup against disasters and regional outages.

You would not create multiple copies of data in several buckets in this scenario. This will cause data duplication and is not a recommended practice. Duplicating data causes a waste of storage and capital. Some ways you can prevent creating duplicate data is by using multi-region buckets and setting object versioning on Cloud Storage correctly. By specifying an object versioning policy, you can maintain a suitable number of copies of data, and you can also use the bucket lock version for protection from the accidental deletion of objects.

You would not set auto stop for virtual machine (VM) instances in this scenario. This is not related to cost optimization for storage. Using auto start and stop for VMs can help you reduce operating costs for Compute Engine. By deactivating VMs during non-business hours, you can make significant savings on costs rather than allowing VMs to run continuously. You can use a serverless system to automate the process of VM shutdowns by using Cloud Pub/Sub.

You would not configure a lifecycle policy for data objects in this scenario. This does not stop the duplication of data. Google Cloud Storage offers various storage classes, which include standard, nearline, coldline, and archival. By moving data to less expensive storage classes, such as the archival class, you can save on costs for storing data that is accessed rarely. By using a lifecycle policy, you can automatically ensure that data objects change their storage class based on conditions you specify, which can be the age of the object.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > Best practices for optimizing your cloud costs](#)

Related Material

Question #47 of 60

Question ID: 1495193

A company that is in the digital publishing domain has approached you to help them find a suitable technology for building a data warehouse that they can use for performing analysis and reporting with scalability that is seamless. Which Google Cloud service should you recommend for this requirement?

A) BigQuery

B) DataFlow

C) Cloud Trace

D) Cloud Storage

Explanation

You should recommend BigQuery for this scenario. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and that can be utilized for analytics. It has features like machine learning (ML), business intelligence (BI), and geospatial analytics ready for use and has a serverless architecture allowing you to use SQL queries for gaining key organizational data. BigQuery has a scalable analysis engine that has a distributed structure. This allows you to perform queries on petabytes of data within minutes.

You would not recommend DataFlow for this scenario because it is not a data warehousing service but a service for performing data processing. DataFlow is a fully managed service that automates the process of provisioning and managing processing resources. It provides the highest possible resource utilization by performing horizontal autoscaling for various worker resources.

You would not recommend Cloud Storage for this scenario because it is an object storage service and not a data warehousing service. Cloud Storage can be used for Object Lifecycle Management (OLM), where data can be configured so that data is automatically moved to lower-cost storage classes when the data meets specified criteria like a period having passed or a new version of the data created. Cloud Storage also provides data redundancy through various locations for storage buckets. This provides fast response times and disaster recovery (DR) options.

You would not recommend Cloud Trace in this scenario as it is used as a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > What is BigQuery?](#)

Related Material

Question #48 of 60

Question ID: 1495926

Which Google Cloud solution can help your company modernize its infrastructure by removing downtime, reducing RTO, and preventing data corruption and loss?

- A) Cloud Run
- B) BigQuery
- C) Cloud Build
- D) Cloud Backup and DR
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Explanation

You should use Cloud Backup and Disaster Recovery (DR) to meet these requirements. This service provides a way to backup workloads from both Google Cloud and on-premises systems for data protection that is consistent as well as centralized. It allows you to have a single system to protect file systems, database instances, and virtual machines (VMs) for both VMware and Compute Engine. Using application-consistent backups that are incremental-forever, Cloud Backup and DR ensures low values for both the recovery point objective (RPO) and recovery time objective (RTO) of a company, ensuring swift recovery of business operations using backups. This is enabled by the ability to instantly mount and use VMware engines and databases from Cloud Backup and DR backups. The incremental-forever system used by Cloud Backup and DR ensures that duplicate data does not need to be stored and results in reduced cost of storage and faster recovery times. This is made possible through changed-block tracking, which reduces storage requirements and speeds up recovery.

You would not use Cloud Run for this scenario. You would use Cloud Run when you need to perform a deployment of containerized applications, which are written in your preferred coding language. Cloud Run provides support for Hypertext Transfer Protocol (HTTP), Hypertext Transfer Protocol Secure (HTTPS), and websockets. You can use Cloud Run for creating application programming interfaces (APIs), applications for processing data, webhooks, and websites. With Cloud Run, code can be packaged into a container and simply deployed without needing infrastructural scaling.

You would not use Cloud Build for this scenario. Cloud Build is used to build, test, and deploy systems using a continuous integration and continuous delivery (CI/CD) platform, which is serverless. It allows a company to integrate its existing legacy systems and tools into a modern build system. This is made possible using third-party applications and customized steps for builds. Cloud Build also has vulnerability scanning features for CI/CD, which prevents the deployment of any images that are unsafe. Some key features of Cloud Build include:

- Fast build time using VMs with fast CPUs provisioned to cache source code and dependencies.
- Deployment automation using pipelines with integrated use of Firebase, Google Kubernetes Engine (GKE), Cloud Functions, and App Engine.
- Multicloud deployment using multilanguage support for builder images used and containerized tasks, which are portable across clouds.
- Strong privacy through Google Cloud security, which allows the use of repositories for private source code inside private networks.

You would not use BigQuery for this scenario. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and that can be utilized for analytics. It has features like machine learning (ML), business intelligence (BI), and geospatial analytics ready for use and has a serverless architecture that allows you to use SQL queries to gain key organizational data. BigQuery has a scalable analysis engine that has a distributed structure and allows you to perform queries on petabytes of data within minutes.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Backup and DR Service](#)

Related Material

Question #49 of 60

Question ID: 1497124

Which of these are characterized as key advantages of moving to a data-driven culture? (Choose two.)

- A) Easy access to analytics data for the business users of a company
- B) Creation of duplicate data for use within an organization
- C) Use of a Zero Trust security model for data sharing
- D) Remove the need for data governance
- E) Less reliance on fully managed services for data management
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Explanation

Easy access to analytics data for the business users of a company and the use of a Zero Trust security model for data sharing are both key advantages of moving to a data-driven culture. A data-driven culture gives data the topmost priority and forms the basis of its decision making for its strategies and teams. A data-driven culture incorporates people, processes, and technology, and ensures that a business:

- operates with trust
- ensures insights are democratized
- increases its agility
- applies business intelligence

Less reliance on fully managed services for data management is not an advantage of moving to a data-driven culture. This is because data can be used efficiently when using more managed integrated systems. This bridges any skill gaps that may exist for users and enables them to operate, scale, and manage data with high levels of agility. This allows for greater user collaboration and removes bottlenecks and silos in an organization.

The creation of duplicate data for use within an organization is not an advantage of moving to a data-driven culture. A data-drive culture seeks to create a single source of reliable data within an organization that can be accessed by users when they need it. This can be achieved by ingestion of data into a data warehouse and building data models with business intelligence systems, such as Looker. This makes relevant data available for all teams without issues related to redundancy and provisioning overhead.

Removing the need for data governance is not an advantage of moving to a data-driven culture. Data governance is an integral part of a data-driven culture and ensures that only the correct users can access relevant data and also ensures that data is safe throughout its lifecycle.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Describe the role of data in digital transformation and the importance of a data-driven culture

References:
[Cloud.google.com > Google's guide to building a data-driven culture](#)

Related Material

Question #50 of 60

Question ID: 1495908

You work as a cloud consultant for a manufacturing firm that wants to use cloud-based business intelligence (BI) tools to identify issues in its production lines. Which of these would you need to do to ensure that your company makes the best use of its BI tools?

- A)** Encourage all users to use open source BI tools
- B)** Opt for an on-premises BI tool for future scalability
- C)** Keep data for BI siloed in different systems
- D) Normalize and clean the data for BI**
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Explanation

You would need to normalize and clean the data for business intelligence (BI). For data to be useful for BI, you need to ensure it is high quality, updated, and cleaned. BI is a process of using technology to collect and study organizational data for making strategic organizational decisions. This involves collecting the data into a storage repository like a data warehouse and using tools to perform data analysis on the stored data. Looker is a BI tool offered by Google Cloud. Some examples of BI applications include analysis of the shopping preferences of customers, costs of company operations, and sales data. BI is also useful for comparing a company's operations versus pre-established standards and baselines. There are three categories of BI tools:

- On-premises – These are used inside an on-premises data center with data warehouses also on-premises. They lack the scalability available in cloud-based BI tools.
- Open source – These are affordable tools and can help save on the costs of infrastructure when used as cloud-based variants. However, they need considerable expertise and coding skills to be used.
- Cloud-based – These are highly effective tools for working with data in high volumes and streaming formats. They are also economical as the technical expertise and infrastructural overhead required for their use is the responsibility of the cloud service provider (CSP).

You would not keep data for BI siloed in different systems as this does not allow users access to all of the data required for BI purposes. Siloed data is distributed across multiple systems and cannot be accessed by other systems due to outdated software or stringent user access permissions. This makes key data needed for BI become inaccessible. Using an Extract, Transform, and Load (ETL) tool can help consolidate the data from across multiple systems needed for BI applications.

You would not encourage all users to use open source BI tools. You need to use managed tools for BI as much as possible because it reduces the complexity of using the tools and ensures that a lack of programming expertise or technical know-how does not become a performance bottleneck in the BI process. Open source BI tools require technical knowledge and may not be suitable for all users inside an organization. You would instead opt for a managed cloud-based BI tool, such as Looker.

You would not opt for an on-premises BI tool for future scalability because on-premises systems are not as scalable as cloud-based solutions. For accommodating future growth and working with massive datasets, you need a cloud-based solution that allows you access to managed data warehouses and BI tools with various features built in for ease of use.

You need to know of some key ways an organization can utilize BI tools, including:

- Reporting of key data for creating strategy and initiatives for the business
- Mining data through massive datasets for identifying trends and patterns
- Providing visualization of data for easy comprehension of complex data
- Analyzing real-time and streaming data, which may be gathered from sources, such as Internet of Things (IoT) sensors and stock market data
- Using predictive modeling and analysis for forecasting future trends using historical data
- Measuring business performance by analyzing sales data and customer feedback

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > What is Business Intelligence?](#)

Related Material

Question #51 of 60

Question ID: 1495226

Your company has a two-tier application with a web-based frontend for interacting with users and an RDBMS backend that stores user data. You need to find a way to gradually move your systems from on-premises to a cloud deployment. How should you implement a tiered hybrid cloud deployment?

- A)** Store data that is restricted by regulations on a public cloud
- B)** Migrate backend applications to a public cloud
- C) Deploy frontend applications to a public cloud**
- D)** Use a private cloud for performing changes to frontends
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Explanation

You should deploy frontend applications to a public cloud. Frontend applications require frequent changes and updates as they face users and are performance centric. By moving frontend applications to a public cloud, you can have an automated continuous integration and continuous delivery (CI/CD) system for efficient deployment of updates to applications.

You would not migrate backend applications to a public cloud. Backend applications can have various system dependencies that are very complex, and the task of migrating these applications to a public cloud can be an involved and challenging process. By focusing on moving the frontend applications to the cloud first, you can perform cloud integration faster and utilize cloud-native services for increased availability and resiliency for the system.

You would not store data that is restricted by regulations on a public cloud. Any data that is subject to regulatory restrictions should ideally be stored in an on-premises environment. This approach ensures greater security of the data.

You would not use a private cloud for performing changes to frontends. This is because a public cloud system offers features like CI/CD for rapid and automated development and updates for applications. Additionally, cloud services, including Cloud Internet of Things (IoT) and Cloud Content Delivery Network (CDN), can be useful for creating user interfaces, data ingestion from IoT sources, and creating application programming interfaces (APIs).

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Hybrid and multi-cloud architecture patterns](#)

Related Material

Question #52 of 60

Question ID: 1495181

You work for a company that is exploring how Google Cloud can allow it to apply its data in new ways. The company wants its applications to be responsive to changing user needs, which can be serviced on demand. What aspect of Google Cloud would be useful in this scenario?

- A) Ingestion and analysis of user data performed in real-time

✓
- B) Validation and cleaning of data done automatically

✗
- C) Storage of data as archives for access over a lengthy period

✗
- D) Removal of overhead related to managing data access

✗

Explanation

Ingestion and analysis of user data performed in real-time is a distinct advantage of using Google Cloud that is relevant to this scenario. To be responsive to changing user needs and preferences, a real-time system is required to collect user data. Gathering and analyzing user data in real-time is done easily on a cloud deployment. Such real-time systems allow applications to make necessary immediate changes to address various user requirements dynamically based on ingested real-time information.

Storage of data as archives for access over a lengthy period is not a relevant advantage in this scenario. Archiving data is a benefit of using the cloud, but it is for data that does not need to be accessed immediately and is not suitable for real-time processing needs. Archiving is a way to keep low-cost storage for infrequently accessed data and meet regulatory and compliance requirements. Archiving using the cloud is not a way to implement real-time user interaction and data requirements.

Validation and cleaning of data done automatically is not a relevant advantage in this scenario. Automatic validation and cleaning of data are useful for applications that need quick and efficient data processing, such as machine learning (ML) applications that better serve users in an efficient and automated manner. The company takes responsibility for the quality and validity of the stored data. This does not represent the kind of real-time ingestion and analysis that is needed in this scenario.

Removal of overhead related to managing data access is not a relevant advantage in this scenario, which requires real-time data processing. Having a centralized Identity and Access Management (IAM) system using user access policies is a key advantage of using a cloud system. When an organization moves to a cloud deployment, it needs to take full responsibility for the access management of its data. That is not the responsibility of the cloud service provider (CSP), but the customer. The CSP looks after the security of the infrastructure that supports the cloud but not of data stored in the cloud by the customer.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Describe the role of data in digital transformation and the importance of a data-driven culture

References:
[Cloud.google.com > Google's guide to building a data-driven culture](#)

Related Material

Question #53 of 60

Question ID: 1495897

Your company has a large on-premises data center that has taken a considerable amount of investment to set up. The company is now looking toward growing its business through the use of a hybrid cloud.

Which of these would be a consequence of using a hybrid cloud solution?

- A) Reduced ROI by the addition of a public cloud

✗
- B) Reduced maintenance of on-premises servers

✗
- C) Increased privacy and compliance for applications

✓
- D) Increased latency for end users

✗

Explanation

You can increase the privacy and compliance of your applications when using a hybrid cloud solution. This is because you can specify the position of your application in relation to the hybrid cloud computing model. This helps you regulate your application, ensuring they fall within compliance requirements and ensure the necessary privacy controls for them.

Reduced maintenance of on-premises servers is not a consequence of using a hybrid cloud solution. While a hybrid cloud solution can allow you to migrate on-premises servers to the cloud and have the cloud provider maintain the hardware for the servers, maintenance for existing on-premises servers will be unaffected by a hybrid cloud solution.

A reduced return on investment (ROI) by the addition of a public cloud does not occur as you increase your computing capabilities through the addition of a public cloud provider to your existing on-premises systems. This can be done without an increase in your existing data center costs leading to an improvement in ROI.

Using a hybrid cloud solution does not increase latency for end users. When you have applications that are distributed at various remote locations, you can greatly benefit from using a hybrid cloud solution. Hybrid computing occurs near the end user's location and leads to reduced levels of latency.

The three main cloud deployment models you should know about include:

- Cloud-based (also known as Cloud native) – In this deployment, applications are fully functional in a cloud and have no on-premises portions.
- On-premises (also known as private) – In this deployment, all the infrastructure and applications that are part of the cloud or virtualization system are running on systems that are inside a company's on-premises data center.
- Hybrid – In this deployment, infrastructure and applications are shared between an on-premises environment and a cloud environment. A hybrid cloud model is used by companies that want to take advantage of the scalability and security offered through a public cloud while maintaining data on-premises for regulatory compliance needs.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why cloud technology is revolutionizing business

References:
[Cloud.google.com > What is a Hybrid Cloud? > What are hybrid cloud benefits?](#)

Related Material

Question #54 of 60

Question ID: 1497139

Your company uses various CRM systems for user experience management. It needs an intelligent system that can extract value from user data to better understand their needs and sentiments. Which Google technology can you use for this requirement?

- A) Looker

B) Memorystore

C) Cloud Translation

D) Contact Center AI
- ✖

✖

✖

✔

Explanation

Contact Center Artificial Intelligence (CCAI) is the appropriate choice for this scenario. CCAI is used to enhance the customer service experience for your customers in existing contact centers by leveraging AI technology. CCAI includes Dialogflow Customer Experience Edition (CX), Dialogflow Essentials (ES), Speaker ID, Text-to-Speech, and Speech-to-Text. The CCAI Platform is a Contact Center as a Service (CCaaS) platform that is AI enabled and created on Google Cloud natively. It is built to be used with Customer Relationship Management (CRM) systems and provides a unified source of data for customer experiences. Contact Center AI Insights enables customers to perform value extraction from their data in contact centers. They can use Contact Center AI Insights to search for key information and perform necessary actions based on that data using a console. Customers can also use Contact Center AI Insights to extract topics, key areas, and sentiments from data using advanced analytics.

Cloud Translation is not the appropriate choice for this scenario. Cloud Translation performs the translation of text between languages using Representational State Transfer (RESTful) APIs. You can use Cloud Translation to translate text from English to Spanish to any other language or vice versa. Using Cloud Translation API, you can provide automatic text translation features for an app or Website.

Looker is not the appropriate choice for this scenario. Looker is a platform for business intelligence that allows you to access intelligent data for making better business decisions. It also can be embedded into other applications to provide embedded analytics. Looker creates alerts for proactively sending timely insights related to various business data. It also has dashboards that are interactive, which allow you to analyze data in real-time through a trusted data model.

Memorystore is not the appropriate choice for this scenario. Memorystore is a fully managed service that provides in-memory storage of data for providing fast data access with less than a millisecond of response time. It does this by the deployment of distributed caches for applications. It helps to migrate the cache layer to the cloud without needing to modify existing application code.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > Contact Center AI](#)

Related Material

Question #55 of 60

Question ID: 1497141

Your development team has limited experience building intelligent systems for image classification based on predefined labels. You need a way to train customized models with the least possible effort so that workloads can be parallelized and less time is spent on manual tasks. Which of these technologies can help you achieve this goal?

- A) Dataproc

B) AutoML

C) Looker

D) Cloud Spanner
- ✖

✔

✖

✖

Explanation

You can use Automated Machine Learning (AutoML) to achieve the necessary objective. AutoML enables you to perform training for high-grade machine learning (ML) models with the least manual effort and knowledge of ML. A user can essentially build a working ML model within a few minutes. Developers can submit their datasets and get to use trained ML models that are built using Google technologies, such as transfer learning and Neural Architecture Search. Neural Architecture Search is Google's technology that provides a way to find, generate, evaluate, and train multiple neural architectures for choosing a solution for a customer automatically. Some key technologies within AutoML include:

- AutoML Natural Language – This allows users to perform categorization of input text into their own predefined labels.
- AutoML Tables – This allows data scientists and developers to build and deploy ML models for structured data at high speeds and scale.
- AutoML Translation – This allows developers to use Google Neural Machine Translation (GNMT) for specific use cases for translation.
- AutoML Video – This allows developers to create ML models for video that are customized and scalable.
- AutoML Vision – This allows developers to train ML models for performing image classification based on custom labels.

The underlying concept behind AutoML involves searching using special algorithms to find solutions that are most optimal for each part of an existing pipeline for ML. This enables a system to be built that automates three main aspects of ML including feature engineering, neural architecture search, and hyperparameter optimization.

You would not use Looker for this scenario. Looker is a platform for business intelligence that allows you to access intelligent data for making better business decisions. It also can be embedded into other applications for providing embedded analytics. Looker can be used to create alerts for proactively sending timely insights related to various business data. It also has dashboards that are interactive, which allow for analyzing data in real-time through a trusted data model.

You would not use Dataproc for this scenario. Dataproc provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, Extract, Transform, and Load (ETL) operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not use Cloud Spanner for this scenario. Cloud Spanner is a relational database service that is fully managed and ideal for critical business applications. You would use Cloud Spanner for provisioning a highly scalable database for online transaction processing (OLTP). This database will have high levels of availability and consistency worldwide.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > AutoML](#)

Related Material

Question #56 of 60

Question ID: 1495934

Your company has embraced a modern application architecture by dividing existing systems into microservices and setting up services management for them. Now, what should you use to manage microservices as APIs?

- A) Apigee**
- B) Dataprep**
- C) BigQuery**
- D) Firestore**
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Explanation
You should use Apigee for managing microservices as application programming interfaces (APIs). Apigee API Management works in tandem with Envoy and Lstio by enabling API management within a microservices stack. Lstio is an open source system for managing and connecting microservices securely. Apigee is used to build, secure, scale, and analyze APIs at any location with support for the complete API lifecycle. It provides strong systems of API control and visibility. With Apigee, you can increase the value of your digital assets by monetizing your API-based products.

You would not use Firestore for this scenario. Firestore is a serverless document database which is scalable and fully managed. It enables fast development for applications for the internet, mobile, and the Internet of Things (IoT), which can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features.

You would not use BigQuery for this scenario. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and that can be utilized for analytics. It has features like machine learning (ML), business intelligence (BI), and geospatial analytics ready for use and has a serverless architecture that allows you to use SQL queries to gain key organizational data. BigQuery has a scalable analysis engine that has a distributed structure and allows you to perform queries on petabytes of data within minutes.

You would not use Dataprep for this scenario. Dataprep is a data service for exploring, cleaning, and preparing data visually for purposes of ML and analysis. It is an intelligent data service and is serverless. This removes the need for users to deploy or manage infrastructure. It also suggests data transformations that are ideal each time a user interface (UI) input is detected, and this removes the need for a user to write code for these applications. Dataprep also detects any anomalies that may exist in the data, including duplicates and missing value data. This helps a user avoid having to assess the quality of data and proceed directly to exploratory data analysis.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Cloud.google.com > Apigee API Management](#)

Related Material

Question #57 of 60

Question ID: 1495284

Your company has adopted the DevOps philosophy of software development. As part of implementing DevOps, which of these will the company need to monitor?

- A) Employee satisfaction**
- B) The time taken for cloud adoption**
- C) The returns made on investments**
- D) System health and reliability**
- 
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Explanation

The system's health and reliability need to be monitored and measured as part of the DevOps philosophy. DevOps is an initiative that is geared toward increasing the speed of delivering software, making services more reliable, and creating ownership that is shared among all stakeholders. DevOps principles are used to enforce security best practices right from the beginning of each development project. Google's DevOps Research and Assessment (DORA) team has found certain key capabilities that an organization can leverage for improved performance and faster software delivery. These capabilities are related to the company's culture, measurement systems, technology, and processes.

Employee satisfaction is not measured as part of the DevOps philosophy. Employee satisfaction depends on various factors, which are not related to the processes and practices that comprise DevOps and its underlying frameworks. DevOps is focused on bringing technology, processes, and people together in a way that facilitates faster software development with greater stability and security.

The returns made on investments are not measured as part of the DevOps philosophy. The measurement of risk and return on investment (ROI) are aspects unrelated to creating an efficient framework for software development that yields stable and secure software delivered swiftly. Monitoring costs and measuring ROI are key functions of business teams and management, and are important for ensuring that the organization's financial performance is in line with the strategy and cost management principles of the company.

The time taken for cloud adoption is not measured as part of the DevOps philosophy. By adopting Google's Cloud Adoption Framework (CAF), a company can make a smooth transition to the cloud based on its current transformational phase. This is not related to creating a DevOps culture and utilizing best practices based on the DevOps system. The CAF is a way for an organization to assess its cloud readiness and find actions to prepare itself for successful cloud adoption. The CAF provides a structure of people, processes, and technologies that are involved in a company that is planning a cloud adoption.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Explain how IT operations need to adapt to thrive in the cloud

References:
[Cloud.google.com > DevOps](#)

Related Material

Question #58 of 60

Question ID: 1497135

Your company is in the process of adopting Google AI technology for its business growth. You are evaluating fairness and security in Google's AI and machine learning differentiators. Which of these is a recommended best practice for Google's stipulated responsible AI?

- A) Perform engagement with multiple users in the development process**
- B) Train models to create automatic inferences based on detected correlations**
- C) Conduct short-term fixes to all issues for swift problem solving**
- D) Keep all the features in your ML model, including redundant ones**

Explanation

Performing engagement with multiple users in the development process is a recommended best practice for Google's stipulated responsible artificial intelligence (AI) framework. This is part of a best practice of utilizing a design approach that is centered on actual users. The effectiveness of an AI system's predictions and decisions can be gauged from the experiences of actual users. Incorporating inputs from a wide range of users and use cases helps enrich the project with a range of perspectives from users. This eventually benefits the users that actually utilize the technology. The system needs to have enough built-in disclosures to users so that a satisfying user experience can be achieved.

Keeping all the features in your machine learning (ML) model, including redundant ones, is not a recommended best practice for Google's stipulated responsible AI. The recommended practice is to use the ML model, which is simple and meets goals for performance without using redundant features. The raw data used by an ML model needs to be analyzed, and if the model uses sensitive data, then that data should be understood within privacy mandates for the data. The data needs to be checked for errors, bias, and training-serving skews to ensure the optimal performance of the model.

Training models to create automatic inferences based on detected correlations is not a recommended best practice for Google's stipulated responsible AI. You need to ensure that the ML model does not make casual inferences based on correlations that it detects. You also need to be aware of the limitations of the model based on its training data's scope and range. Communicating these limitations to users can help with getting useful feedback about the AI system from users.

Conducting short-term fixes to all issues for swift problem solving is not a recommended best practice for Google's stipulated responsible AI. Short-term fixes may save time but do not offer solutions that are beneficial in the long-term. For this reason, it is important to create a balance between using short-term fixes and long-term solutions. Also, you need to compare the candidate ML model with the deployed version to identify the ways they differ, and to understand how the update will alter the user experience and system performance of the AI application.

Some additional best practices include using metrics like user feedback, false negatives and positives, product health reports, and system performance data for assessing monitoring and training of the ML model. Also, testing the system with both integration tests as well as unit tests to ensure that the performance of the AI system is as expected. This involves studying how the individual components of the ML model work with other components of the application. Also, continuous monitoring of the model needs to be done to ensure the model incorporates feedback from the real-world and user experiences.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Ai.google/responsibilities > Responsible AI practices](#)

Related Material

Question #59 of 60

Question ID: 1495220

Your company currently has a legacy on-premises data center, which it uses to provide online shopping services to its customers. Which of these features will require the company to modernize its IT infrastructure with Google Cloud?

- A) Deploying global applications that are fault tolerant and highly available**
- B) Increasing the physical security of the data center**
- C) Performing vertical scaling of data center infrastructure**
- D) Connecting legacy systems to the Internet using VPNs**

Explanation

Deploying global applications that are fault tolerant and highly available requires modernizing IT infrastructure with Google Cloud. A legacy infrastructure cannot deliver a global fault tolerant infrastructure at the level and scalability of a cloud deployment. Google Cloud provides a global infrastructure that consists of regions and zones. This gives you the flexibility to choose how you want to deploy your applications for the best levels of reliability and availability. It also provides the least latency possible for end users through the use of edge locations. Regions are geographical areas that are independent. Each region consists of zones, which are areas of deployment of Google resources inside of a region. A zone can be considered as a single failure domain, which implies that for an application to have fault tolerance and high availability, it needs to be deployed across multiple zones. The services of Google's infrastructure are available globally and have round the clock maintenance and upgrades performed. Google's global infrastructure makes the process of managing multi-region deployments simple and provides flexible load balancing features.

Performing vertical scaling of data center infrastructure does not require a cloud deployment. Vertical scaling involves increasing the computing power of existing machines and can be done within a data center itself. This is not a cloud exclusive feature. However, a cloud deployment would be more appropriate if there was a requirement for computing resources to be adjusted dynamically as per changing user demand. This is a concept called elasticity, which ensures that the optimum amount of cloud resources is allocated for handling workloads.

Connecting legacy systems to the Internet using virtual private networks (VPNs) does not require a cloud deployment. For connecting computer systems to the Internet, you need an Internet connection through an Internet Service Provider (ISP), and a VPN connection through a VPN provider. However, modernizing the infrastructure with Google Cloud does have the advantage of improving the security of the existing systems through strong identity and access management as well as network protection tools to safeguard against Internet-based malware.

Increasing the physical security of the data center does not require a cloud deployment. The physical security of the data center is the responsibility of the company and needs to be done by the company itself. The security measures will include security barriers, such as walls and fencing; and monitoring systems, such as cameras and sensors. Security measures might also include guards and card- or token-based access control.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Geography and regions](#)

Related Material

Question #60 of 60

Question ID: 1497133

You need to implement a system for ensuring data quality in prediction accuracy for deployed ML models. How would you do this?

- A) Use continuous integration and delivery for ML models**

✔
- B) Use known data for ML models without regular updating**

✘
- C) Maintain accurate ML models without retraining**

✘
- D) Manually update each ML model with fresh data**

✘

Explanation

You would use continuous integration and delivery for machine learning (ML) models for ensuring data quality in prediction accuracy for the deployed ML models. ML needs to have some quality control measures implemented for the best results. These measures include processes to evaluate models after training and to validate them pre-deployment. Additionally, there needs to be logging and monitoring systems in place so that the systems health can be checked and notifications issued in the event of any anomaly. When continuous integration/continuous delivery (CI/CD) is applied to ML, it involves validation of ML models, data schemas, and code as part of CI, and deployment of ML models and continuous testing (CT) pipelines as part of CD. To create a production system for ML, you would use ML Metadata, Container Registry Cloud Build, and Model Registry services.

You would not manually update each ML model with fresh data. This is not a suitable approach because manual updating can lead to errors and is not a feasible solution in an enterprise that may have a massive number of deployed ML models that need regular updates of their data as well as implementation of new ML concepts. An automated pipeline-based approach would be ideal for updating ML models. This approach is used in transformational maturity, which involves orchestration and automation of ML and data pipelines end-to-end.

You would not use known data for ML models without regular updating. This is because ML models need to be updated due to changes in the data that characterizes their operating environment. Failure to update ML models with fresh data can lead to ML models failing in deployment. Using CT as part of CD ensures that the ML models are fully tested and validated pre-deployment and re-trained regularly.

You would not maintain accurate ML models without retraining because ML models need to be trained continuously to have the most accurate predictions. This is to ensure that the models are accurate and can respond to real world data correctly. Deploying a CT pipeline as part of a larger CD system ensures that the ML models are retrained regularly. This ensures that all ML models currently deployed into production are updated with fresh data, which can capture trending patterns as well as applications of the latest ML concepts.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Services.google.com > Google Cloud's AI Adoption Framework](#)

Related Material